

**Abundance and Run Timing of Adult Fall Chum
Salmon in the Teedriinjik (Chandalar) River,
Yukon Flats National Wildlife Refuge, Alaska, 2017**

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Cover: Aerial view of the Teedriinjik (Chandalar) River sonar camp, August 2014

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Abundance and Run Timing of Adult Fall Chum Salmon in the Teedriinjik (Chandalar) River, Yukon Flats National Wildlife Refuge, Alaska, 2017

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Abstract

During 2017, Dual Frequency Identification Sonar (DIDSON) was used to assess the abundance of adult fall Chum Salmon *Oncorhynchus keta* in the Teedriinjik River (formerly known as the Chandalar River before its name was changed in 2015), a tributary of the Yukon River. Sonar operations began on August 8 and continued until September 26.

In 2017, 30-minute samples of each hour were counted and those counts were expanded to estimate hourly passage. A total of 1,061.9 hours of acoustic data was analyzed, resulting in 225,090 upriver swimming fish enumerated. After expanding sample counts and adjusting for missed time, the fall Chum Salmon passage estimate for 2017 was 483,833 fish, with a 95% confidence interval of 482,227–485,439 fish. This is the second highest passage estimate since the project began in 1987. The upriver swimming fall Chum Salmon passage estimate was 1,706 fish on the first day of counting and 8,160 fish on the last day of counting. The first-quarter point of the run occurred on September 8, the mid-point on September 15, and the third-quarter point on September 19. Overall, run timing was slightly later than average.

This passage estimate is conservative because it only includes fish that passed during the dates of sonar operation and within the ensonified portion of the river. Most fish were shore-oriented, and few were observed near the outer range limits of the ensonified zones suggesting that most fish were within the detection range of the DIDSON.

Introduction

Accurate salmon escapement counts on Yukon River tributaries are important for informing and assessing annual harvest management decisions, predicting run strength based on brood year returns, and monitoring long-term population trends. Weirs, counting towers, mark-recapture programs, ground surveys, and hydroacoustics are used to obtain escapement estimates of specific Yukon River Salmon stocks (Bergstrom et al. 2001).

The Yukon River drainage encompasses 854,700 km² and is among the largest producers of wild Chinook Salmon *Oncorhynchus tshawytscha* and Chum Salmon *O. keta* in North America (Daum and Osborne 1995). The salmon resources of this unique river support important subsistence and commercial fisheries throughout the drainage. The U.S. Fish and Wildlife Service (USFWS), through Section 302 of the Alaska National Interest Lands Conservation Act and the Yukon River Salmon Agreement, has a responsibility to ensure that salmon populations within federal conservation units are conserved in their natural diversity, subsistence opportunities are maintained, and international treaty obligations are met. Estimating spawning escapements for major salmon stocks in the drainage is essential for addressing these mandates.

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The Teedriinjik River (formerly known as the Chandalar River before its name was changed in 2015) is a tributary of the Yukon River. The fall Chum Salmon stock in the Teedriinjik River is one of the largest stocks in the Yukon River drainage (JTC 2016) and is an important resource for wildlife and subsistence users.

The use of fixed-location hydroacoustics to count migrating salmon in Alaska began during the early 1960s and has provided counts in rivers where limited visibility or sample volume precluded other sampling techniques (Gaudet 1990). The USFWS conducted a 5-year study from 1986 to 1990 using fixed-location Bendix side scanning sonar salmon counters to enumerate adult fall Chum Salmon in the Teedriinjik River. The annual Bendix sonar counts of fall Chum Salmon during that period averaged 58,457 fish, with a range of 33,619–78,631 fish (Appendix A). These early Bendix sonar salmon counters were not acoustically calibrated, used factory-set echo-counting criteria to determine fish counts, had limited acoustic range (<33 m), and could not determine the direction of target travel (upriver or downriver). Due to these technological limitations, it is suspected that, overall, the Bendix sonar yielded low estimates of fall Chum Salmon passage on the Teedriinjik River.

A study was initiated in 1994 to reassess the Teedriinjik River fall Chum Salmon population status using split-beam sonar technology. This sonar technology was acoustically calibrated, had user-defined echo-tracking techniques to count fish, and had an extended acoustic range (>100 m). The split-beam sonar also provided three-dimensional positioning for each returning echo, which allowed the determination of direction of travel and swimming behavior for each passing target (Daum and Osborne 1998). Operations during 1994 were used to develop site-specific procedures and methods, evaluate site characteristics, and describe possible data collection biases (Daum and Osborne 1995). In 1995, daily and seasonal estimates of fall Chum Salmon passage were calculated post season and *in situ* target strength evaluations were collected (Daum and Osborne 1996). Fall Chum Salmon enumeration with the split-beam sonar continued through 2006.

Experimentation to evaluate DIDSON (Dual Frequency Identification Sonar) technology from Sound Metrics (Lake Forest Park, Washington), for enumeration of fall Chum Salmon in the Teedriinjik River was initiated in 2004 and continued through 2006. DIDSON offers several advantages over the previous sonar technologies used on the Teedriinjik River. These include deployment over a wider range of site conditions, production of a more easily interpreted visual image, reduction of training requirements for technicians due to operation and image interpretation that is more intuitive, easier setup and deployment, and the potential for increased capacity for species determination under some conditions. The primary limitations of DIDSON, relative to split-beam sonar, include limited range capabilities, lack of vertical position data, and larger data files requiring large hard drives to store or archive data. During the evaluation period, up to three DIDSONs were set up at different locations, both adjacent to the split-beam sonar, and at independent locations. Conclusions from these evaluations indicate that DIDSON is well suited to enumerate fall Chum Salmon on the Teedriinjik River (Melegari 2008). Therefore, DIDSON has been used on the Teedriinjik River since 2007.

Objectives of the Teedriinjik River sonar project have consistently been to: (1) provide daily in-season passage estimates of Teedriinjik River fall Chum Salmon to fishery managers; (2) estimate annual passage of fall Chum Salmon; and (3) describe annual variability in run size and timing. Since 1996, the project has provided daily in-season passage estimates to managers and an estimate of annual passage post season. Sonar passage estimates from 1995 to 2017 averaged 208,213 fall Chum Salmon and ranged from 65,894 to 496,484 fall Chum Salmon (Appendix A).

Several years of gill net and beach seine sampling and observations with underwater video cameras have shown that, during sonar operations, the number of fish of other species in the same size range as Chum Salmon is inconsequential relative to the number of Chum Salmon present (Daum and Osborne 1995, 1996, 1998; Osborne and Melegari 2006; Melegari and Osborne 2007; Melegari 2011, 2012). Therefore, all upriver swimming fish are assumed to be Chum Salmon.

Additionally, carcass sampling for age, sex, and length data has been conducted at fall Chum Salmon spawning grounds upstream of the community of Venetie on the Teedriinjik River during some years (Appendix B); however, funding was unavailable for this work in 2017.

Study Area

The Teedriinjik River is a fifth-order tributary of the Yukon River that drains part of the southern slopes of the Brooks Range. The Teedriinjik River drainage consists of three major branches (Figure 1). These branches are the Teedriinjik River, which includes the main branch from the mouth up to and including what was formerly known as the North Fork Chandalar River prior to 2015, the Ch'idriinjik River, which was formerly known as the Middle Fork Chandalar River prior to 2015, and, the East Fork Chandalar River (name was not changed in 2015). The sonar site is located on the lower Teedriinjik River, below any of the other two branches or any notable tributaries (Figure 1). The Teedriinjik River enters the Yukon River approximately 32 km downriver of Fort Yukon. Principal water sources include rainfall, snowmelt, and to a lesser extent, melt water from small glaciers, and perennial springs (Craig and Wells 1975). Summer water turbidity is highly variable, depending on rainfall. The region has a continental subarctic climate characterized by some of the largest temperature extremes in the state, -59°C to 38°C (USFWS 1987). Precipitation ranges from 15 to 33 cm annually with the greatest amount falling between May and September. The river is typically ice-free by early June and freeze-up typically occurs in early October.

The lower 19 km of the Teedriinjik River is influenced by a series of slough systems connected to the Yukon River. River banks are typically steep and covered with overhanging vegetation and downed trees caused by active bank erosion. Gravel bars are generally absent in this area and the bottom substrate is primarily sand and silt. Water velocities are generally less than 0.75 m/s. From approximately 21.0 to 22.5 km upriver from its confluence with the Yukon River, the Teedriinjik River is confined to a single channel with steep cut-banks alternating with large gravel bars. Substrate in this area primarily ranges from small gravel to cobble with some sand/silt in slow current areas. Upriver from this area, the river becomes braided with many islands and multiple channels. The sonar study area is in a remote location 21.5 km upriver from the Teedriinjik River's confluence with the Yukon River (Figure 2) and is described in more detail by Daum et al. (1992).

The left bank (facing downriver) DIDSON site has a bottom slope of approximately 5° out to approximately 40 m where it flattens out (Figure 3). On the right bank site, the bottom slopes at approximately 7° out to approximately 27 m before it flattens out. Substrate on both banks consists mainly of large gravel. Overall river width at the site varied with water levels from approximately 143 m to 158 m during 2017.

Methods

Water Quality Monitoring

A staff gauge was installed upstream of the right bank sonar to measure daily water levels, which were calibrated using a benchmark established on the right bank in 1989. Water temperatures were taken daily at approximately 1200 hours using a glass thermometer. Additionally, a YSI model 6920V2 sonde was deployed to collect temperature (°C), conductivity (µS/cm), dissolved oxygen (mg/L and % saturation), pH, and turbidity (NTU) data every 15 minutes. The sonde was deployed on August 10 and collected data through September 26. During 2017 pH readings decreased to unreasonably low values during the season and post-season calibrations and evaluation of the pH probe indicated that the probe failed. Therefore, all pH readings for 2017 were discarded.

Site Selection and Sonar Deployment

A sonar deployment site for each bank was selected from several cross-sectional river profiles of the area, developed using a Lowrance (Tulsa, Oklahoma) HDS-7 depth sounder–GPS.

Requirements for site selection included: (1) single channel; (2) uniform non-turbulent flow; (3) gradually sloping bottom gradient without sudden inflections; (4) absence of structure or debris that could impede fish detection; (5) location downriver from known salmon spawning areas; and (6) active fish migration past the site (no milling behavior).

The Sound Metrics DIDSON system is a high frequency 12° X 29° multiple beam sonar (Belcher et al. 2001, 2002). Standard and long-range models are currently in use. The standard DIDSON operates at frequencies of 1.8 or 1.1 MHz and has an effective range for confidently enumerating fall Chum Salmon on the Teedriinjik River of approximately 30 m, based on detection of known targets drifted through the sonar beam, and on analysis of fish data. The long-range model operates at frequencies of 1.2 MHz or 700 KHz with an effective range of approximately 60 m. DIDSON specifications are available in the DIDSON operation manual V5.25 (Sound Metrics Corp. 2010). The DIDSON units were deployed at fixed locations in the river and communicated with laptop computers for control and data management.

The long-range DIDSON was deployed on the left bank and the standard DIDSON was used on the right bank. Both DIDSON models were operated in the low frequency mode (1.1 MHz for the standard range and 700 KHz for the long range). Partial weirs, constructed with T-posts and 2 in x 4 in (5.08 cm x 10.16 cm) welded wire fence, were installed approximately 1 m downriver of each DIDSON to direct migrating fish through the beams. The left bank DIDSON had a window start setting of 0.83 m, and a window length setting of 40.01 m (the window settings determine the range within which the DIDSON acquires data). The right bank had a window start setting of 0.83 m, and a window length setting of 20.01 m. The left bank recorded at one frame per second, and the right bank recorded at six frames per second. The recording rate on the left bank was limited by the slower wireless network used to transmit data across the river, and by the longer window setting.

The DIDSON units were oriented perpendicular to river flow and mounted to aluminum frames with brackets allowing manual adjustments to vertical and horizontal aim. The aim was adjusted by verifying the detection of targets (liter plastic bottle half filled with steel shot) placed on the river bottom at varying ranges within the ensonified area and drifted through the ensonified area from a boat.

A wireless network was installed for the left bank so all DIDSON communications, data acquisition, and analysis could occur at a single data tent located on the right bank next to the camp. This remote communications network consisted of two D-Link® DWL-2100AP wireless access points, one connected to the DIDSON on the left bank, and the other, connected to the receiving computer on the right bank. A D-Link® ANT24-1800 outdoor directional panel antenna was attached to each access point using an outdoor low loss radio frequency cable.

Sonar Data Collection and Analysis

In the data tent, a wired network was set up for each DIDSON to facilitate data collection and analysis. Each DIDSON was networked with two laptop computers and a 4-terabyte network attached storage (NAS) drive. One computer was used to control and communicate with the DIDSON and save the collected data to files on the NAS. The second computer was used to analyze the data and manage data files. Additional backup copies of the count data and summary data files were saved to USB flash drives at the end of each day.

The sonar systems were operated continuously 24 hours per day, except during intermittent periods for maintenance, repairs, aim adjustments, relocating the DIDSON as water levels changed, or during periods of high water and/or heavy debris loads that would impede the safe operation of the sonar equipment. The collected data were saved to files at 30-minute intervals. Data were analyzed using the DIDSON control and display software (version 5.25.32; Sound Metrics Corp. 2010). Data files were examined in the echogram view and when a potential target was encountered it was further evaluated by reviewing that section of data in the normal image view to verify the target was a fish and to determine direction of travel. Data from counted files were then exported to ASCII text files, which were compiled and summarized using a Microsoft Excel Visual Basic for Applications macro developed by the author.

Because most fish present during sonar operations are Chum Salmon, apportioning sonar counts to Chum Salmon was based on relative target size as each fish passed through the ensonified field. All upstream swimming fish that appeared large enough to be a Chum Salmon on the DIDSON were assumed to be Chum Salmon. While actual length measurements from the DIDSON are not precise, relative fish sizes can be observed, and fish that were obviously smaller than Chum Salmon were not counted. Conversely, fish larger than typical Chum Salmon are very uncommon during sonar operations. Additionally, previous years of beach seining, gill netting, and underwater video monitoring all indicate that the majority of fish present during sonar operations that are in the same size range as Chum Salmon are Chum Salmon (Daum and Osborne 1995, 1996, 1998; Osborne and Melegari 2006; Melegari and Osborne 2007; Melegari 2011, 2012).

During 2017 a sampling protocol, in which 30-minute fish counts from each hour were expanded to obtain hourly fish passage estimates, was used to reduce crew size and save associated project costs. Prior to first initiating this sampling protocol in 2014, a comparison of full 24-hour counts and estimates from simulated 30-minute sample counts was conducted using data from 2011 to 2013 (Melegari 2015). This comparison showed that the sampling protocol produced estimates with acceptable accuracy and precision. Additionally, we compared mean hourly passage rates (the proportion of the daily passage that passed during a given hour) calculated from the 30-minute sample counts with those calculated from full 24-hour counts. In all cases, the values calculated from the samples were equivalent to those calculated from the complete counts (author's unpublished data). The first 30 minutes of each hour was used for the sample. If the first 30-minute data file of an hour was missing or incomplete, then the second 30-minute data

file for that hour was used. If neither 30-minute data file in an hour contained at least 28 minutes of continuous data, that hour was treated as a missing hour.

Thirty-minute sample counts were expanded to obtain hourly passage estimates using the formula:

$$E_h = (60 / T_h) \cdot C_h \quad (1)$$

Where E_h = estimated hourly fish passage for hour h , T_h = number of minutes sampled in hour h , and C_h = upriver count during the sampled time during hour h .

Fish counts for missing hours were estimated from seasonal mean hourly passage percent. Hourly passage percent (proportion (%)) of the total daily count passing during a given hour were determined for all hours in each day from days when all 30-minute sample periods were fully sampled. These hourly passage rates were expressed as a proportion (%) of the daily count so high-passage days did not bias results. Then, mean hourly passage percent for the season was calculated for each hour. During the season, missing hours were estimated using the averaged mean hourly passage percent from all previous seasons. During post-season analysis, missing hours were re-estimated using seasonal mean hourly passage percent for that year. Estimated fish counts for missing hours were calculated, using:

$$E_d = \sum R_{di} / (100 - \sum R_{di}) \cdot T_d \quad (2)$$

Where E_d = estimated upriver fish count for missing hours in day d , R_{di} = mean hourly passage percent for each missing hour i in day d , and T_d = adjusted upriver fish count for non-missing hours in day d .

Daily upriver fish passage estimates for each bank were calculated by summing all hourly passage estimates for that day. For the season, total fish passage was calculated by summing all daily estimates. Hourly fish passage rates for each bank were plotted for the season and examined for diel patterns. Range distributions of fish targets were evaluated to assess the likelihood of fish passing beyond the detection range of the DIDSON.

During the last day of operation, September 26, counting continued through 1200 hours on both banks. These counts were expanded for the full day using the mean hourly passage percent as previously described.

For calculating the variance of the seasonal estimate from the sampling, a sample observation consisted of the sum of counts from both banks. If either bank had a missing count during an hour then we treated that hour as a missed sample observation. Variance ($\hat{V}$) of the seasonal estimates was calculated using the V5 sample variance estimator suggested by Reynolds et al. (2007):

$$\hat{V} = (1 - f)(1/n) \sum_{j=5}^n c_j^2 / (3.5(n - 4)) \quad (3)$$

Where f is the proportion of possible observations that were actually observed, $c_j = y_j/2 - y_{j-1} + y_{j-2} - y_{j-3} + y_{j-4}/2$, and y_j is the j th observation of a systematic sample of n observations.

A 95% confidence interval for the total seasonal estimate was calculated using the normal interval estimator also recommended by Reynolds et al. (2007):

$$95\%CI = \hat{Y} \pm 1.96 \sqrt{\hat{V}(\hat{Y})} \quad (4)$$

Where $\hat{Y}$ = seasonal passage estimate and $\hat{V}$ = variance of the seasonal passage estimate.

Results

Water Quality Monitoring

Calibrated water levels ranged from approximately 1.9 to 3.5 m and averaged 2.7 m during the 2017 season (Figure 4; Appendix C). River width at the sonar deployment site ranged from approximately 143 to 158 m. Daily water temperatures taken with the glass thermometer ranged from 4°C to 18°C and averaged 9.0°C. Water temperatures, recorded with the YSI sonde during sonar operations, ranged from 3.9°C to 16.8°C and averaged 8.8°C. Staff gauge and Sonde water quality data are presented in Appendix C.

Site Selection and Sonar Deployment

Several cross-sectional profiles were recorded on each bank near the previous deployment locations and the DIDSONs were deployed where the bottom profiles were considered best for counting fish with the DIDSONs. The same approximate locations on both banks have been used since 2007 when operations switched to DIDSON, given that minimal changes in physical conditions have been observed at these locations between years.

Sonar Data Collection and Analysis

Sonar operations began on August 8 and continued through September 26 in 2017. Acoustic data were collected for 1,052 of the scheduled 1,200 30-minute sample periods. A total of 1,061.9 hours of acoustic data were analyzed. Approximately 104 hours were missed between August 30 and September 3 due to high water and heavy debris loads that interrupted counting. Additionally, approximately 34 hours were missed due to repositioning the DIDSONs as water levels changed during the season and due to equipment maintenance or malfunctions. A total of 234,802 fish were counted and upriver swimming fish accounted for 95.9% (225,090) of the total (Table 1). Upriver swimming fish counts were 98,189 for the left bank and 126,901 for the right bank.

After expanding sample counts, and adjusting for missed sampling time, the estimated fall Chum Salmon passage for 2017 was 483,833 fish (Table 2; Appendix D), with a 95% confidence interval of 482,227 to 485,439 fish. The 95% confidence interval only includes the variance associated with subsampling and does not include any other sources of error. Passage during the high-water event (August 31–September 2) was estimated by linear interpolation using August 30 and September 3 as “known” data points. The left bank estimate was 217,079 fall Chum Salmon accounting for 45% of the total. The right bank estimate was 266,754 fall Chum Salmon accounting for 55% of the total. The adjusted count was 1,706 upriver swimming fall Chum Salmon on the first day of sonar operation (0.4% of the total), and 8,160 upriver swimming fall Chum Salmon on the final day of counting (1.7% of the total). Peak daily passage occurred on September 15 (Figure 5; Table 2). The first-quarter point of the run occurred on September 8, the mid-point on September 15, and the third-quarter point on September 19.

Mean hourly passage rates (number of fish estimated to have passed during each hour expressed as a proportion of the daily estimate) of upriver swimming fall Chum Salmon showed weak diel patterns on both banks (Figure 6). On the left bank, the pattern displayed slightly higher counts during late night and early morning (2000–0600 hours). On the right bank, the highest counts occurred during late afternoon and evening (1500–2000 hours) and during mid to late morning (0600–1100 hours).

Range distribution data indicates that upriver swimming fall Chum Salmon were shore-oriented and most were within the range of acoustic detection for both banks. More than 50 % of upriver swimming fall Chum Salmon were within the first 3 m on the left bank and the first 4 m on the right bank (Figure 7); and more than 95% were within 10 m on the left bank, and 13 m on the right bank. Downriver swimming fish, while also generally shore-oriented, were slightly more dispersed across the full detection range of the DIDSONs on both banks.

Discussion

Water Quality Monitoring

During 2017, water levels were relatively high throughout the majority of the season, especially during the second half of the season (Figure 4). The average water level during 2017 was one of the highest recorded during project operations, however only a few days of counting were missed due to high water levels.

For a second consecutive year, the pH probe on the sonde deployed at the sonar project failed during the season. Similar to 2016, the pH readings began to drop slowly soon after deployment and continued to steadily drop to unlikely readings. During post-season calibration and evaluation, the pH probe generated voltage readings that were outside of values specified by YSI for acceptable probe operation. The sonde used in 2017 was a different sonde than was used in 2016, and all probes functioned properly at a different project earlier in the season, and during preseason calibrations.

Site Selection and Sonar Deployment

Similar to most years, little to no changes in bottom profiles were detected at the deployment sites in 2017. The absence of substantial changes in bottom profiles since 2007 has allowed for consistent deployment, data collection, and analysis.

Data were collected at one frame per second on the left bank with the long range DIDSON and the wireless communications network. One frame per second was considered sufficient to capture fall Chum Salmon migrating upriver past the site. This is supported by the data, in which nearly all fish were captured in several frames. If substantial numbers of fish were being missed because the frame rate was too low, then we would expect more fish to be “nearly missed” or captured in only one or two frames.

Sonar Data Collection and Analysis

Run timing was later than average in 2017. The first-quarter point (September 8) occurred 8 days later than average (1995-2014, excluding 2009 when the project terminated early); the mid-point (September 15) occurred 6 days later than average; and the third-quarter point (September 19) occurred 3 days later than average. This is consistent with preliminary data for overall Yukon River fall Chum Salmon run timing, where all assessment projects averaged 2 days later (J. Estensen and B. Borba, 2017 Alaska Department of Fish and Game, Division of Commercial

Fisheries, 2017 Yukon Area Fall Season Summary News Release, unpublished data). The estimated passage of 8,160 fall Chum Salmon on the last day of sampling was 1.7% of the cumulative, indicating some fish passage continued after enumeration stopped. Thus, the actual quarter points of the run may be slightly later than calculated.

The 2017 passage estimate of 483,833 fall Chum Salmon is 248% of the historical average (Figure 9), and is the second highest passage estimate recorded at the project. This is consistent with preliminary data from other Yukon River fall Chum Salmon assessment projects during 2017, which all indicated runs were above normal (J. Estensen and B. Borba, 2017 News Release, unpublished data).

The observed diel patterns in upriver fish passage during 2017 were similar to, though weaker than, patterns seen during previous years (Daum and Osborne 1998; Melegari and Osborne 2007; Figure 6). During most years, the left bank exhibits a strong diel pattern, whereas the right bank generally exhibits a weaker or no diel pattern.

The right bank accounted for 55% of the total passage during 2017, this is the lowest percentage recorded during the project. Higher counts on the right bank have been observed during all years of the project (Appendix A), with an average of 71% of total passage occurring on the right bank. Since switching to DIDSON, enumeration has been able to continue during higher water than was possible with the split-beam sonar. Observations during these higher water levels have shown a tendency for an increased proportion of fish passage on the left bank during the high water. Thus, it seems likely that the higher than usual water levels throughout 2017 could account for the lower than usual proportion of fish passage on the right bank.

Fish range data collected with the DIDSONs were similar to previous years and suggests that most upriver swimming fish passing the sonar site were within the ensonified zone. Upriver swimming fish were found close to shore with relatively few fish found near the range limits of acoustic detection. This shore orientation is consistent with previous behavioral observations of upriver-migrating fall Chum Salmon on the Teedriinjik (Daum and Osborne 1998; Osborne and Melegari 2006), Sheenjek (Barton 1995), and main-stem Yukon rivers (Johnston et al. 1993; Lozori and McDougall 2016).

Unlike split-beam sonar, DIDSON does not obtain vertical fish position data. However, the larger vertical angle of the DIDSON's beams (12° vs. the 2.1° and 4.8° of the split-beam previously used on the Teedriinjik River) reduces the potential of fish passing above or below the beams. This is further supported by the DIDSON data, in which surface waves were usually detected on windy days, and the river bottom was normally visible throughout the range.

Conclusions

The DIDSONs performed well overall. In 2017, there was a short period when high water accompanied by heavy debris required removal of the DIDSONs from the water from August 31 through September 2. These missed days were estimated by linear interpolation. Although this method adds an undetermined amount of error to the seasonal upriver passage estimate, it is currently the most suitable method under the circumstances. Additionally, since the time missed included only 104 hours of sampling, and those hours occurred when daily passage was low relative to daily passage during the peak of the run, the effect on the seasonal passage estimate is presumed to be inconsequential.

The 30-minute sampling protocol used in 2017 worked well and produced a small variance estimate and tight confidence intervals for the estimated passage. During simulations of this sampling protocol using data from the 2011 to 2013 seasons, 95% confidence intervals for the seasonal passage estimate included the actual value from the full count for all 3 years (Melegari 2015).

The Teedriinjik River sonar passage estimate is likely a conservative estimate because it does not include fish that passed before or after sonar operation. Additionally, while Chum Salmon are generally shore and bottom oriented during migration, it is probable that some fish passed undetected outside of the ensonified zone.

Annual sonar enumeration of fall Chum Salmon in the Teedriinjik River is a vital component for effectively managing the complex mixed-stock subsistence and commercial fisheries in the Yukon River. The Teedriinjik River fall Chum Salmon stock is crucial to the total Yukon River fall Chum Salmon run and is important to users throughout the drainage. Daily in-season counts and post-season passage estimates provide important escapement information to managers and users of this resource, allowing for better-informed management decisions and evaluation of previous management actions. This project is a key component in assessing the lower river abundance estimate proportioned by mixed-stock genetic analysis. Additionally, this project has provided accurate population status and trend data for 23 years. These time series data will become more important as stressors such as climate change, disease, selective harvest, and overall demand on the fisheries and resources in the Yukon River drainage continue to increase.

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Table 1. — Hydroacoustic data collected via DIDSON at the Teedriinjik (Chandalar) River, Alaska, 2017.

Date	Left bank			Right bank			Combined		
	Sample time (h)	Upriver count	Downriver count	Sample time (h)	Upriver count	Downriver count	Sample time (h)	Upriver count	Downriver count
8-Aug	8.00	219	0	8.00	360	11	16.00	579	11
9-Aug	12.00	208	10	12.00	470	4	24.00	678	14
10-Aug	12.00	146	14	11.94	346	8	23.94	492	22
11-Aug	12.00	77	2	12.00	269	4	24.00	346	6
12-Aug	12.00	99	6	12.00	198	9	24.00	297	15
13-Aug	12.00	79	8	12.00	181	10	24.00	260	18
14-Aug	12.00	112	0	12.00	209	5	24.00	321	5
15-Aug	11.00	169	12	10.95	268	15	21.95	437	27
16-Aug	8.00	308	4	9.00	230	7	17.00	538	11
17-Aug	11.97	595	2	12.00	572	10	23.97	1,167	12
18-Aug	12.00	608	0	12.00	623	1	24.00	1,231	1
19-Aug	12.00	734	6	12.00	703	2	24.00	1,437	8
20-Aug	12.00	830	4	12.00	641	0	24.00	1,471	4
21-Aug	12.00	799	2	12.00	573	3	24.00	1,372	5
22-Aug	12.00	742	0	12.00	718	1	24.00	1,460	1
23-Aug	11.50	797	8	12.00	1,042	7	23.50	1,839	15
24-Aug	12.00	881	0	12.00	1,052	2	24.00	1,933	2
25-Aug	12.00	1,041	0	12.00	959	10	24.00	2,000	10
26-Aug	11.50	988	5	11.00	1,266	14	22.50	2,254	19
27-Aug	12.00	1,190	3	12.00	1,635	9	24.00	2,825	12
28-Aug	12.00	1,537	2	12.00	1,832	10	24.00	3,369	12
29-Aug	12.00	1,563	2	12.00	1,552	15	24.00	3,115	17
30-Aug	5.00	1,105	5	3.50	136	15	8.50	1,241	20
31-Aug ^a									
1-Sep ^a									
2-Sep ^a									
3-Sep	3.50	541	1	4.00	551	0	7.50	1,092	1
4-Sep	12.00	1,755	6	12.00	1,379	38	24.00	3,134	44
5-Sep	12.00	1,521	13	12.00	1,361	60	24.00	2,882	73
6-Sep	11.50	1,146	4	11.50	1,403	74	23.00	2,549	78
7-Sep	12.00	1,102	10	12.00	1,694	55	24.00	2,796	65
8-Sep	12.00	1,638	19	12.00	1,970	191	24.00	3,608	210
9-Sep	12.00	2,911	12	12.00	2,254	103	24.00	5,165	115
10-Sep	12.00	3,352	2	12.00	2,654	38	24.00	6,006	40
11-Sep	12.00	4,290	8	12.00	4,031	63	24.00	8,321	71
12-Sep	12.00	4,815	12	12.00	5,352	122	24.00	10,167	134
13-Sep	12.00	4,570	15	12.00	7,124	125	24.00	11,694	140
14-Sep	12.00	5,014	16	12.00	10,757	64	24.00	15,771	80
15-Sep	12.00	5,862	57	12.00	11,245	226	24.00	17,107	283
16-Sep	12.00	6,581	37	12.00	10,039	205	24.00	16,620	242
17-Sep	12.00	6,636	31	12.00	7,156	206	24.00	13,792	237
18-Sep	12.00	6,689	49	12.00	6,230	222	24.00	12,919	271
19-Sep	12.00	5,204	72	12.00	6,828	379	24.00	12,032	451
20-Sep	12.00	4,553	140	12.00	5,890	583	24.00	10,443	723
21-Sep	12.00	4,865	601	12.00	5,656	2,176	24.00	10,521	2,777
22-Sep	12.00	3,938	207	12.00	5,533	466	24.00	9,471	673
23-Sep	11.00	2,177	115	12.00	4,351	211	23.00	6,528	326
24-Sep	12.00	1,994	83	12.00	3,696	248	24.00	5,690	331
25-Sep	11.00	1,394	236	12.00	2,696	876	23.00	4,090	1,112
26-Sep	6.00	814	172	6.00	1,216	796	12.00	2,030	968
Totals	529.97	98,189	2,013	531.89	126,901	7,699	1,061.86	225,090	9,712

^a Sonar not operating due to high water.

Table 2. — Daily fall Chum Salmon upriver passage estimates at the Teedriinjik (Chandalar) River, Alaska, 2017.

Date	Left bank	Right bank	Combined	Cumulative	Cumulative %
8-Aug	703	1,003	1,706	1,706	0.35
9-Aug	416	940	1,356	3,062	0.63
10-Aug	292	695	987	4,049	0.84
11-Aug	154	538	692	4,741	0.98
12-Aug	198	396	594	5,335	1.10
13-Aug	158	362	520	5,855	1.21
14-Aug	224	418	642	6,497	1.34
15-Aug	362	594	956	7,453	1.54
16-Aug	859	626	1,485	8,938	1.85
17-Aug	1,195	1,144	2,339	11,277	2.33
18-Aug	1,216	1,246	2,462	13,739	2.84
19-Aug	1,468	1,406	2,874	16,613	3.43
20-Aug	1,660	1,282	2,942	19,555	4.04
21-Aug	1,598	1,146	2,744	22,299	4.61
22-Aug	1,484	1,436	2,920	25,219	5.21
23-Aug	1,651	2,084	3,735	28,954	5.98
24-Aug	1,762	2,104	3,866	32,820	6.78
25-Aug	2,082	1,918	4,000	36,820	7.61
26-Aug	2,040	2,781	4,821	41,641	8.61
27-Aug	2,380	3,270	5,650	47,291	9.77
28-Aug	3,074	3,664	6,738	54,029	11.17
29-Aug	3,126	3,104	6,230	60,259	12.45
30-Aug	4,846	1,152	5,998	66,257	13.69
31-Aug	4,537 ^a	1,654 ^a	6,191	72,448	14.97
1-Sep	4,228 ^a	2,156 ^a	6,384	78,831	16.29
2-Sep	3,919 ^a	2,657 ^a	6,576	85,408	17.65
3-Sep	3,610	3,159	6,769	92,177	19.05
4-Sep	3,510	2,758	6,268	98,445	20.35
5-Sep	3,042	2,722	5,764	104,209	21.54
6-Sep	2,398	2,930	5,328	109,537	22.64
7-Sep	2,204	3,388	5,592	115,129	23.80
8-Sep	3,276	3,940	7,216	122,345	25.29
9-Sep	5,822	4,508	10,330	132,675	27.42
10-Sep	6,704	5,308	12,012	144,687	29.90
11-Sep	8,580	8,062	16,642	161,329	33.34
12-Sep	9,630	10,704	20,334	181,663	37.55
13-Sep	9,140	14,248	23,388	205,051	42.38
14-Sep	10,028	21,514	31,542	236,593	48.90
15-Sep	11,724	22,490	34,214	270,807	55.97
16-Sep	13,162	20,078	33,240	304,047	62.84
17-Sep	13,272	14,312	27,584	331,631	68.54
18-Sep	13,378	12,460	25,838	357,469	73.88
19-Sep	10,408	13,656	24,064	381,533	78.86
20-Sep	9,106	11,780	20,886	402,419	83.17
21-Sep	9,730	11,312	21,042	423,461	87.52
22-Sep	7,876	11,066	18,942	442,403	91.44
23-Sep	4,767	8,702	13,469	455,872	94.22
24-Sep	3,988	7,392	11,380	467,252	96.57
25-Sep	3,029	5,392	8,421	475,673	98.31
26-Sep	3,063	5,097	8,160	483,833	100.00
Totals	217,079	266,754	483,833		

^a Counts estimated by linear interpolation using August 30 and September 3 as “known” points

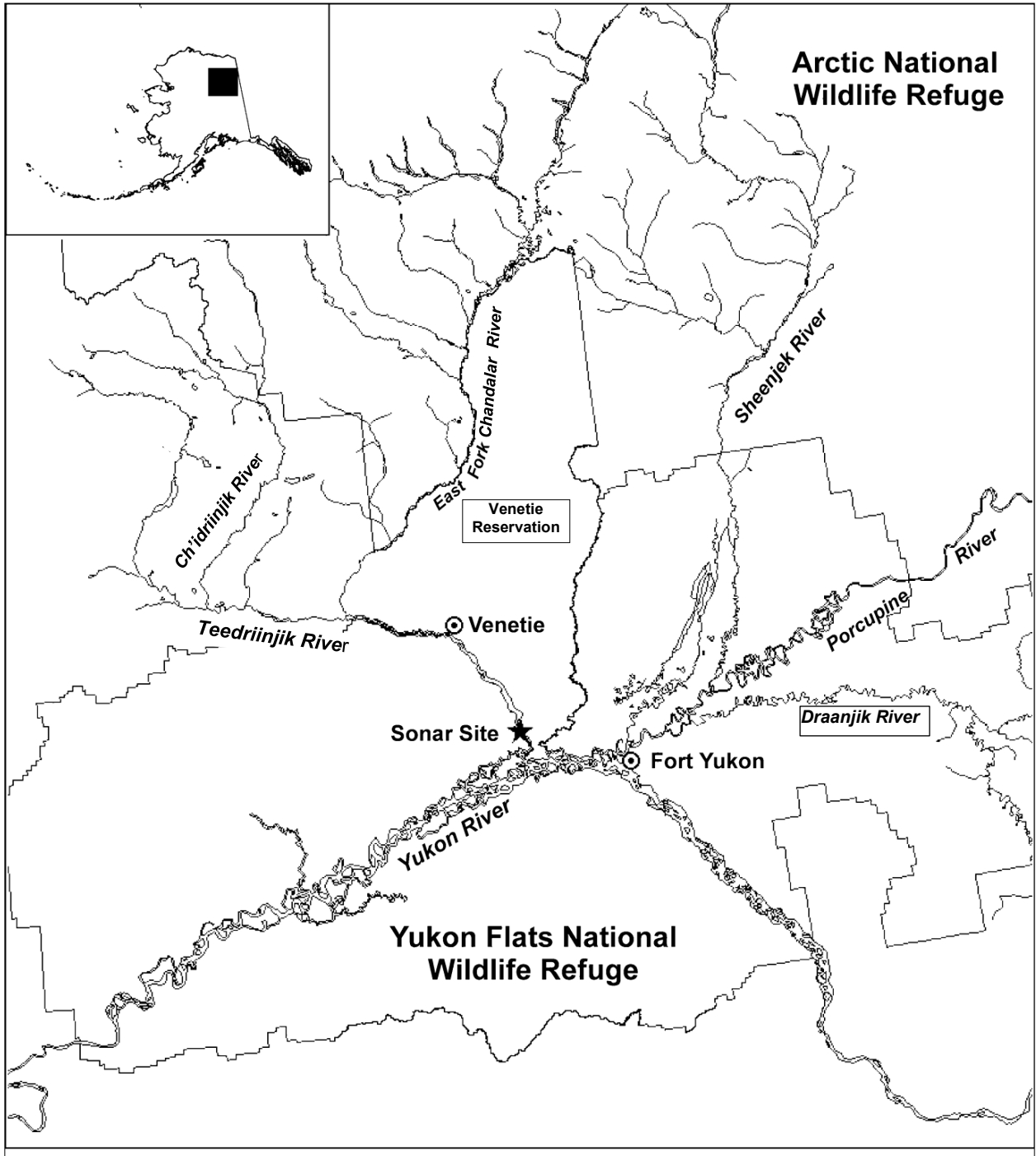


Figure 1. — Teedriinjik (Chandalar) River sonar site and major tributaries of the Yukon River near the U.S.-Canada border.

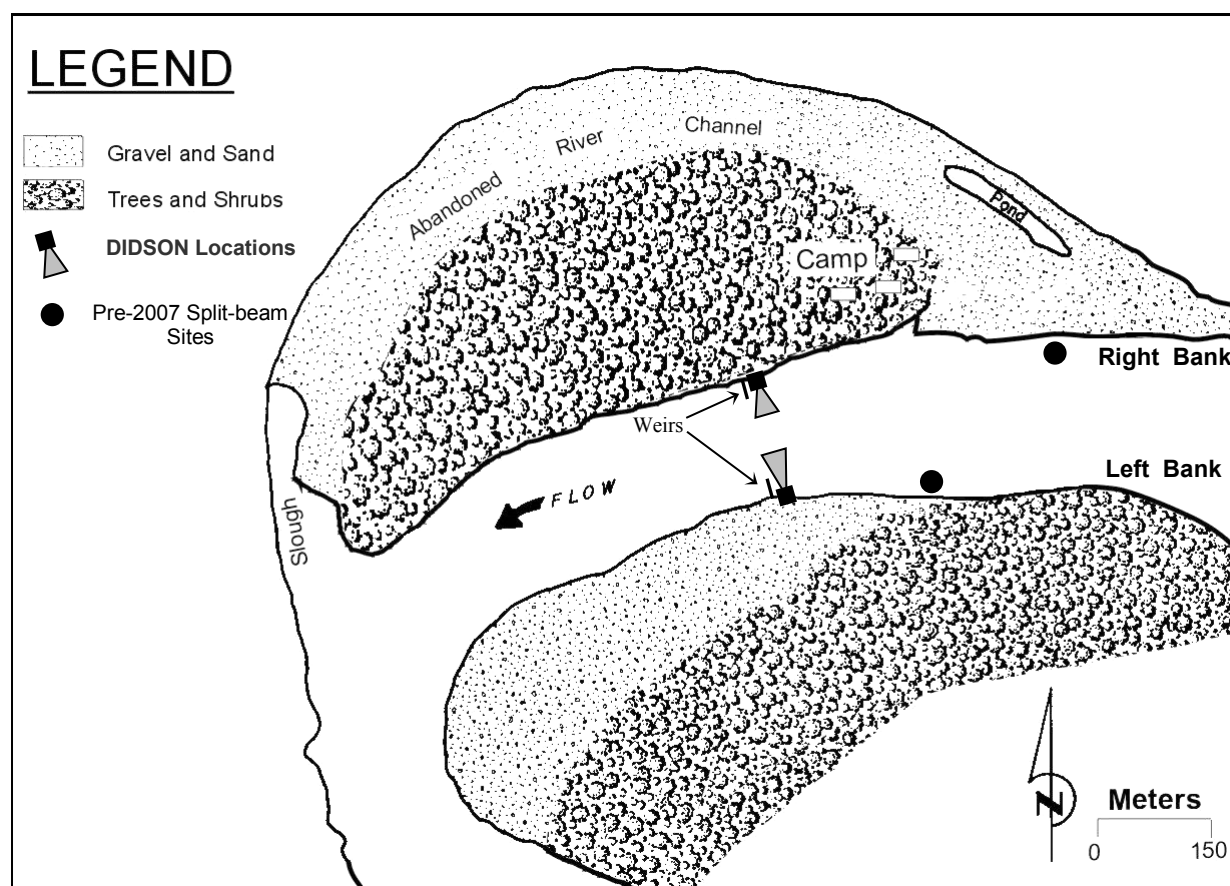


Figure 2. — Site map of Teedriinjik (Chandalar) River sonar facilities.

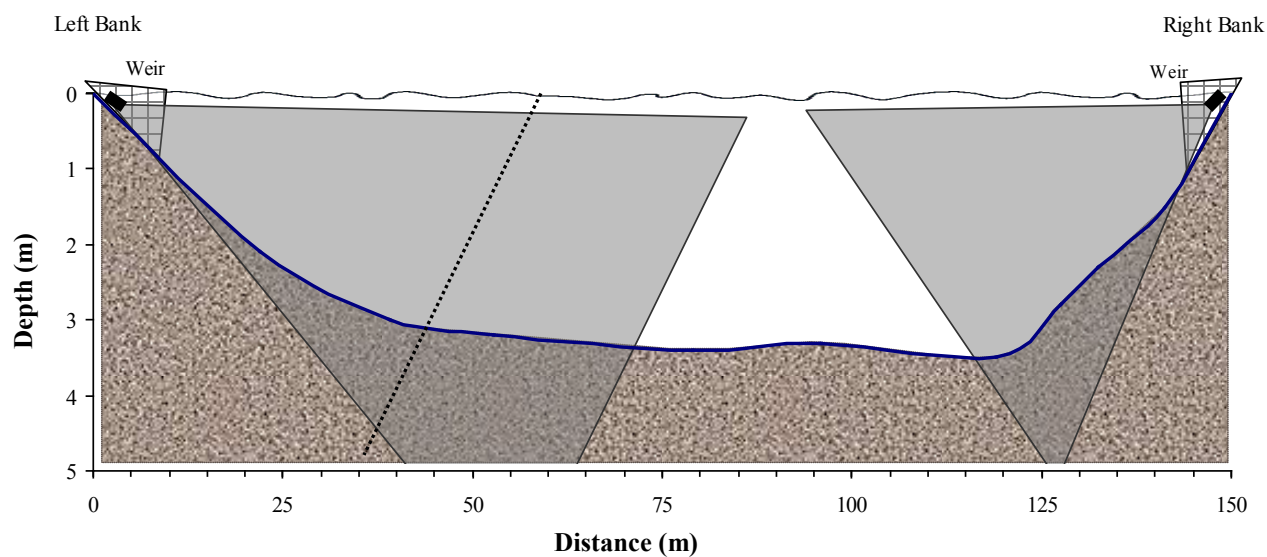


Figure 3. — River channel profile and approximated ensonified zones during 2009 for the left and right bank sonar sites, Teedriinjik (Chandalar) River. Dotted line on left bank indicates the approximate limit of the ensonified zone with the range reduced to 40 m. Little change has occurred in channel profile from 2008 to 2017. Note: different axis scales are used to enhance readability.

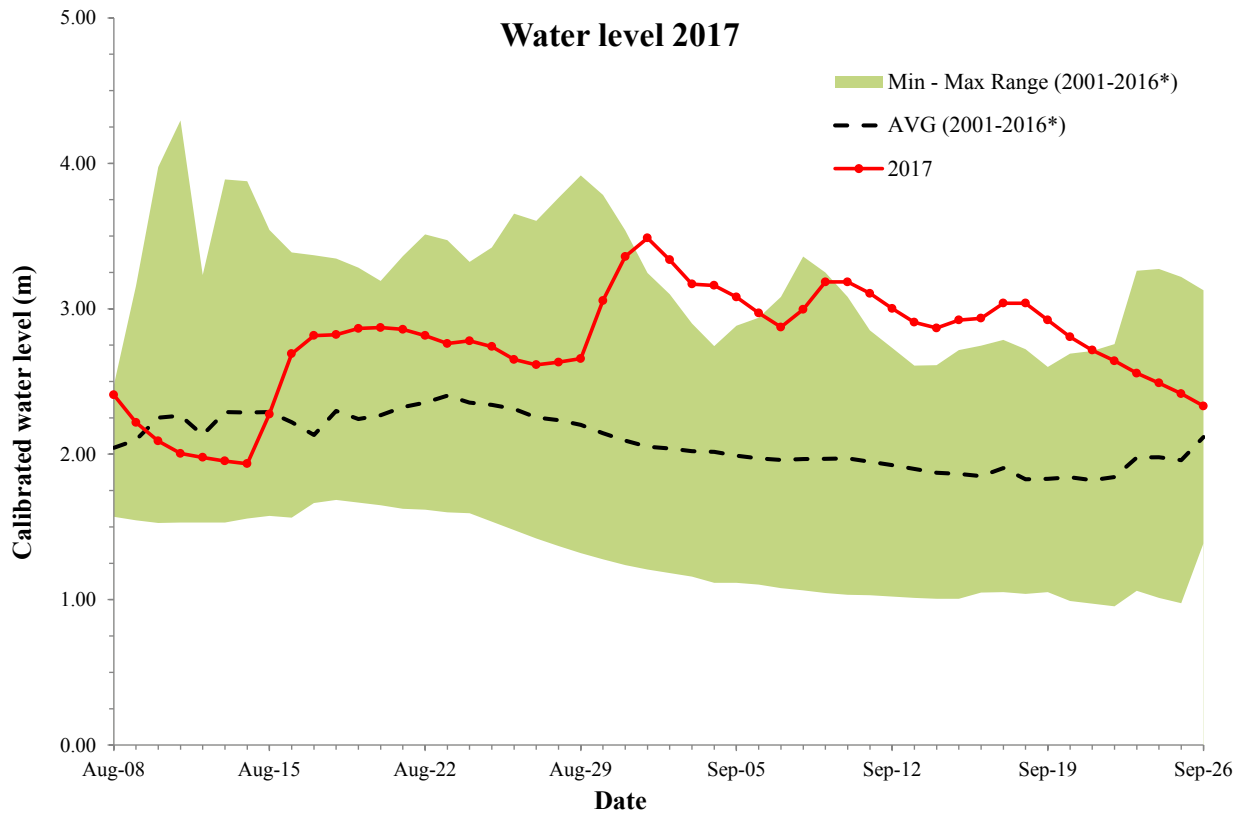


Figure 4 — Daily calibrated water levels from the Teedriinjik (Chandalar) River sonar site, 2017. Green shaded area represents historical range of minimum and maximum water levels. Historical min – max ranges and averages do not include data from 2003, 2005, 2006, and 2009.

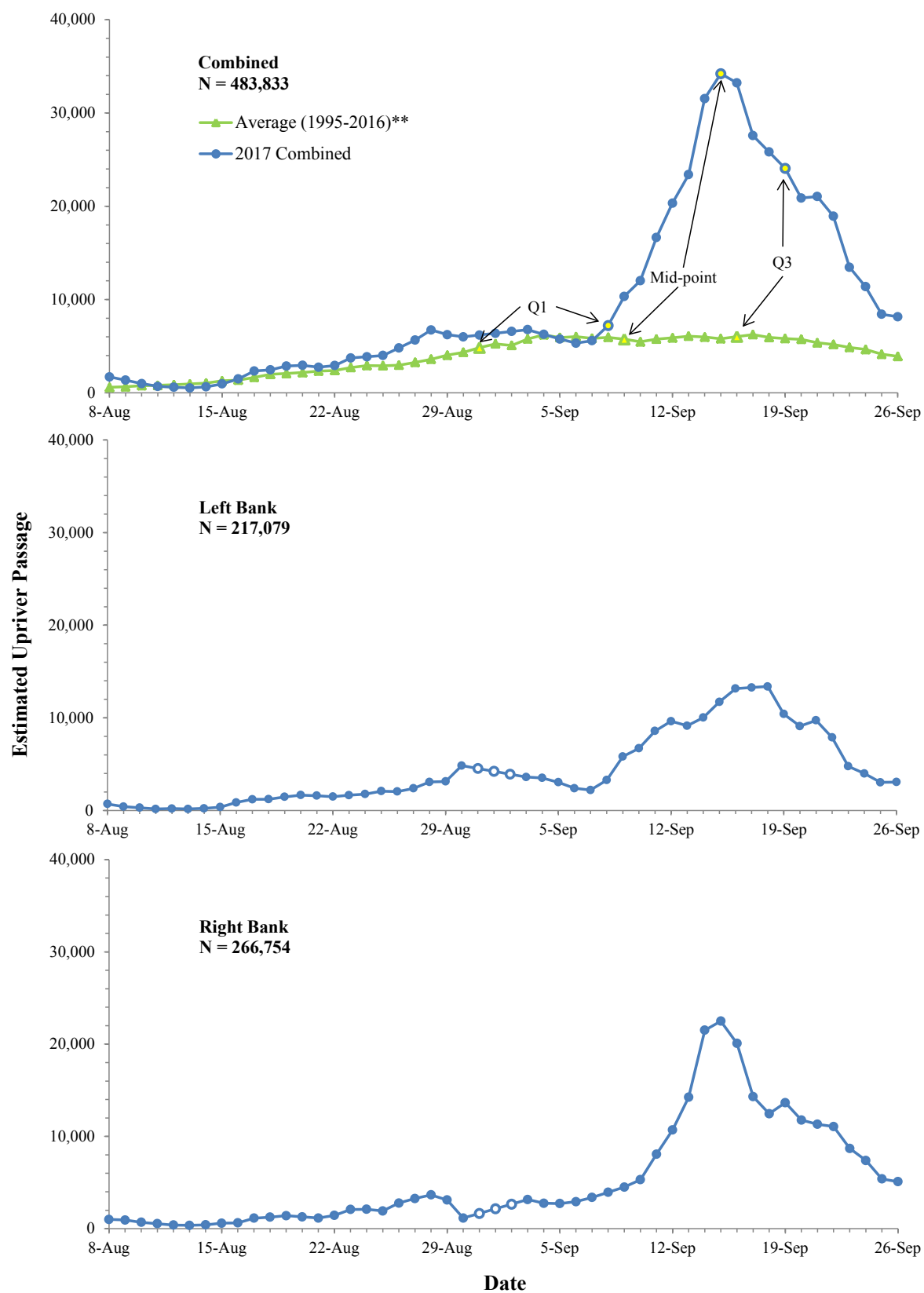


Figure 5. — Estimated passage of upriver swimming fall Chum Salmon by bank and combined, Teedriinjik (Chandalar) River, 2017. Highlighted points in the top graph indicate the 1st-quarter, mid, and 3rd-quarter points of passage. *Passage estimates from 8/31 to 9/2 for both banks (unfilled points) were estimated by linear interpolation due to high water interrupting sonar operations. **Average does not include data from 2009 because the project ended early before the majority of the run normally passes.

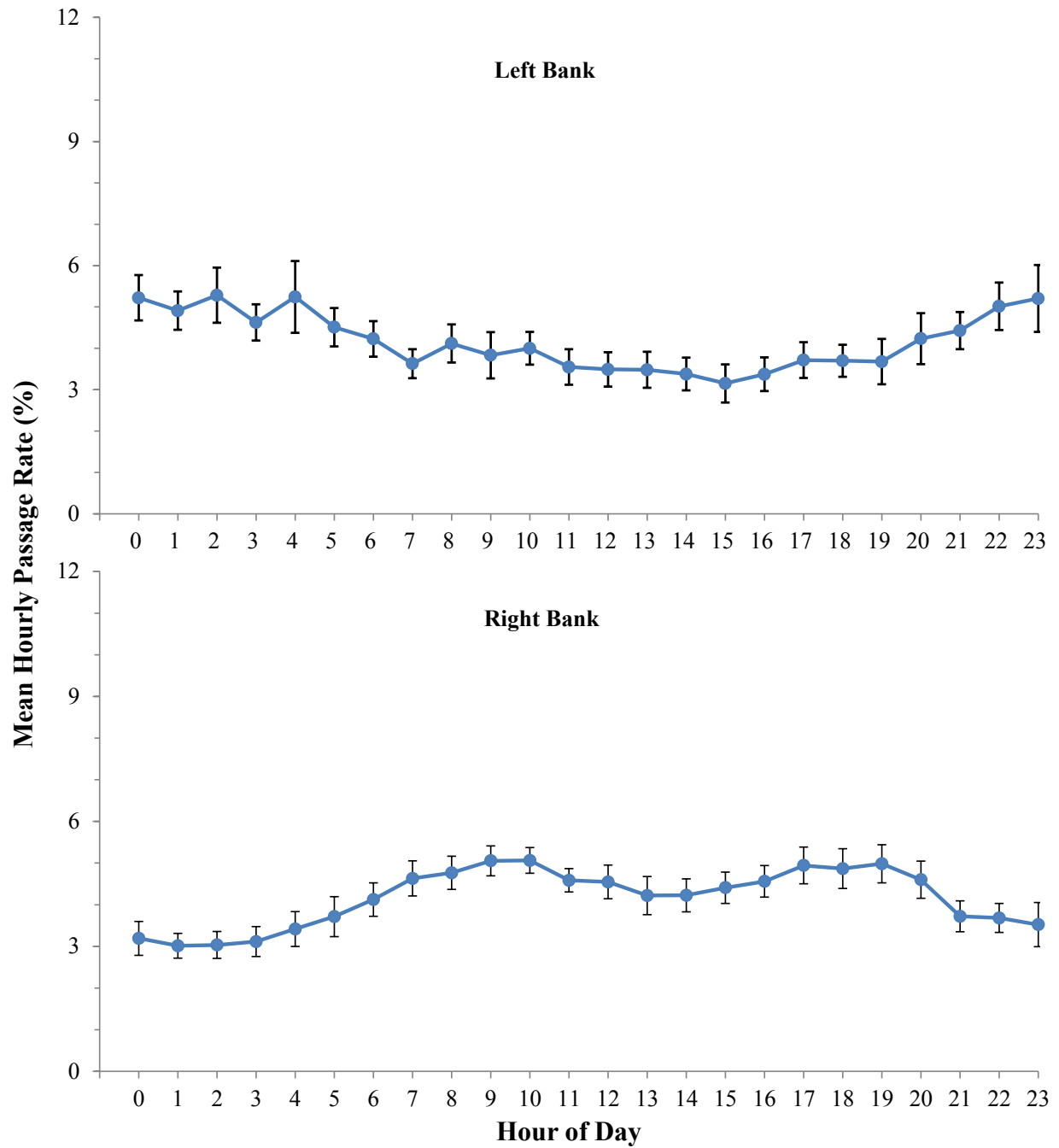


Figure 6. — Mean (± 2 SE) hourly passage rate (expressed as a percent of the total daily estimate) of upriver swimming fall Chum Salmon, Teedriinjik (Chandalar) River, 2017. Results include 36 days of complete sampling (12 hours/day) on the left bank and 39 days on the right bank.

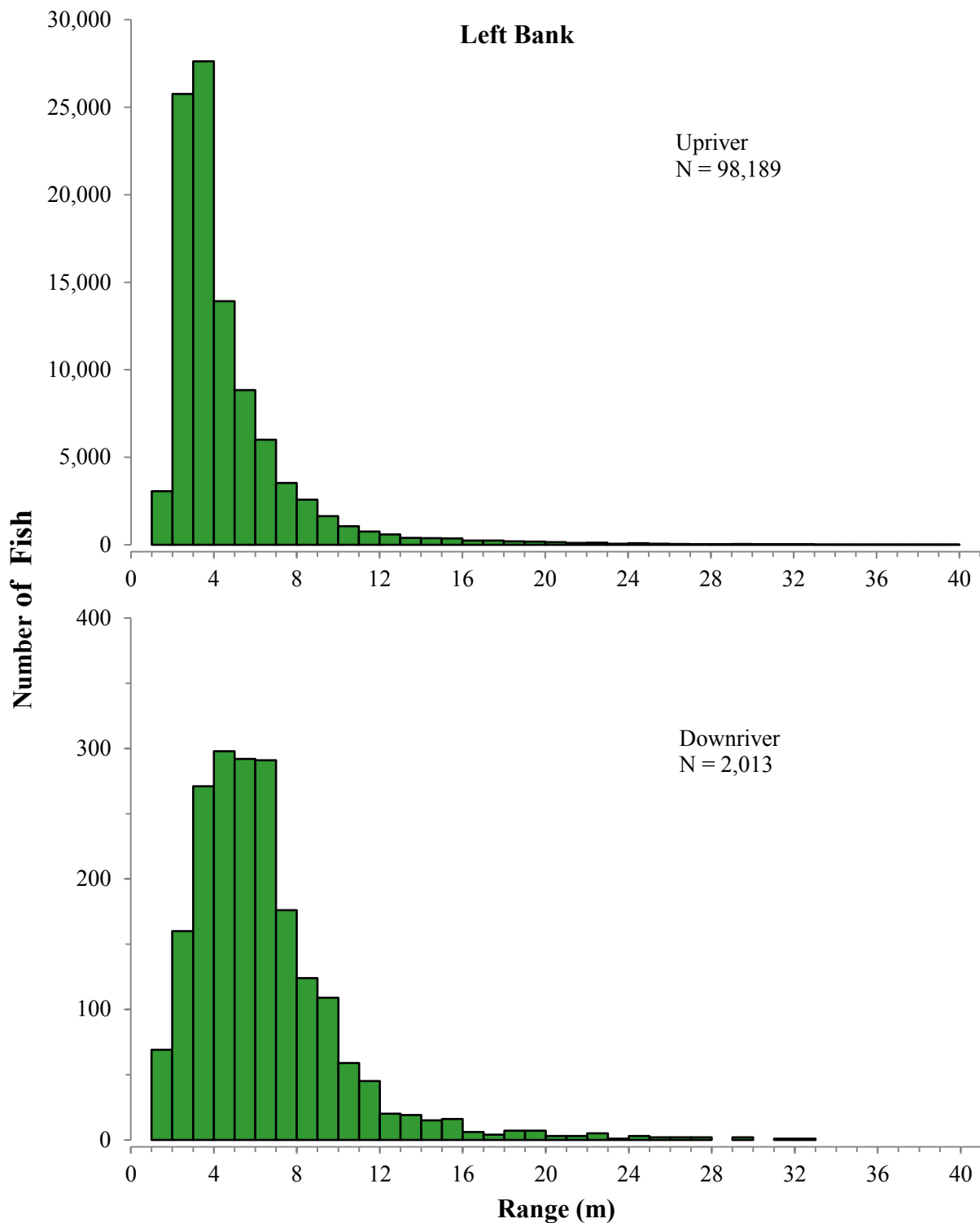


Figure 7. — Range (horizontal distance from DIDSON) distribution of upriver and downriver swimming Chum Salmon from hydroacoustic data collected on the left bank Teedriinjik (Chandalar) River, August 8 to September 26, 2017. Note different Y-axis scales.

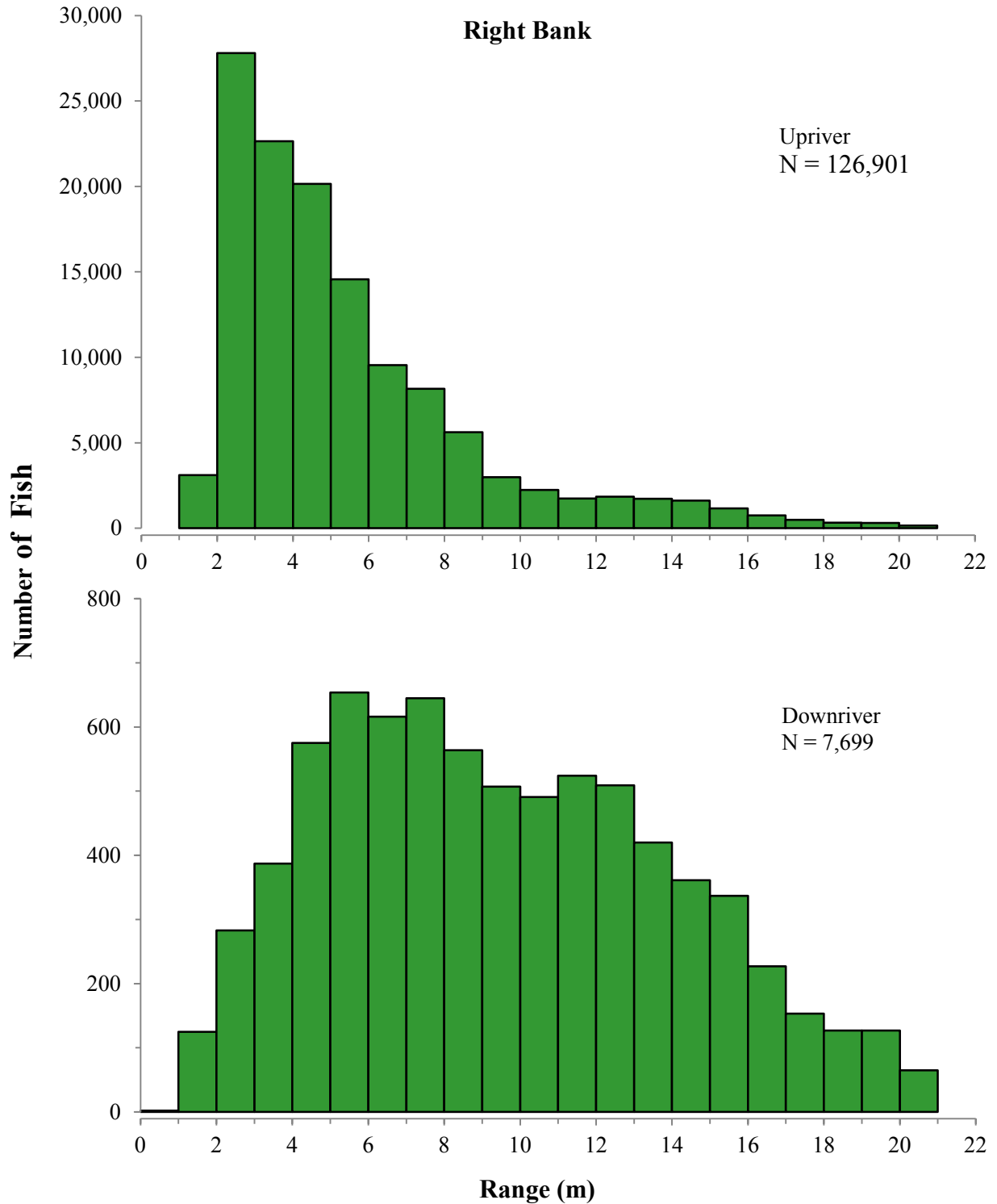


Figure 8. — Range (horizontal distance from DIDSON) distribution of upriver and downriver swimming Chum Salmon from hydroacoustic data collected on the right bank Teedriinjik (Chandalar) River, August 8 to September 26, 2017. Note different Y-axis scales.

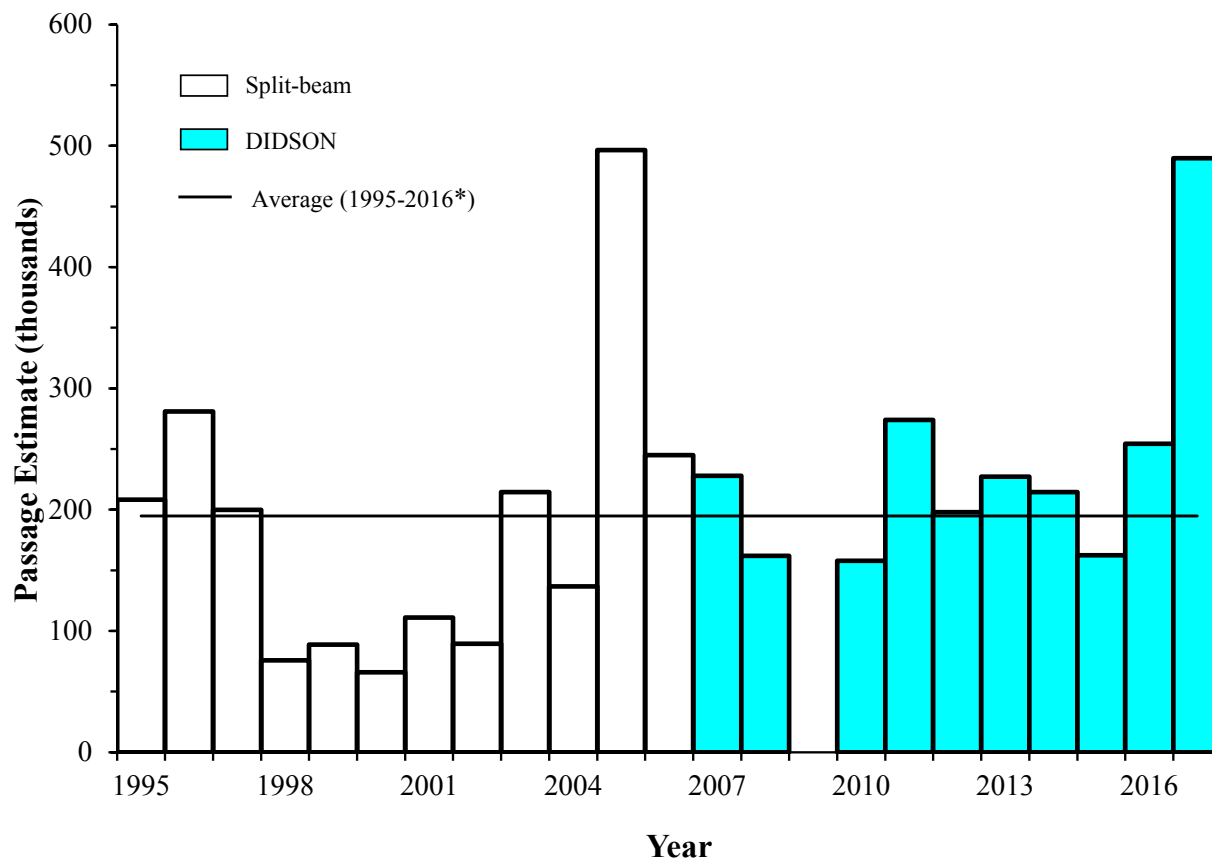


Figure 9 — Annual passage estimates (in thousands of fish) of fall Chum Salmon from sonar counts on the Teedriinjik (Chandalar) River, 1995-2017. The horizontal line indicates the average of 1995 to 2016 passage estimates. *Average does not include data from 2009 because the project ended early before the majority of the run normally passes.

Appendix A. Historical Passage

Table A.1. — Historical fall Chum Salmon passage estimates from sonar counts on the Teedriinjik (Chandalar) River, Alaska.

Year	Sonar type	Passage estimate		
		Left bank	Right bank	Combined
1987	Bendix	36,089	16,327	52,416
1988	Bendix	20,516	13,103	33,619
1989	Bendix	36,495	32,666	69,161
1990	Bendix	24,635	53,996	78,631
1995 ^a	Split-beam	116,074	164,925	280,999
1996	Split-beam	75,630	132,540	208,170
1997	Split-beam	65,471	134,403	199,874
1998	Split-beam	31,676	44,135	75,811
1999	Split-beam	38,091	50,571	88,662
2000	Split-beam	16,420	49,474	65,894
2001	Split-beam	20,299	90,672	110,971
2002	Split-beam	24,188	65,392	89,580
2003	Split-beam	68,825	145,591	214,416
2004	Split-beam	29,851	106,852	136,703
2005	Split-beam	159,937	336,547	496,484
2006	Split-beam	63,123	181,967	245,090
2007	DIDSON	31,193	196,862	228,055
2008	DIDSON	22,261	139,763	162,024
2009 ^b	DIDSON	1,314	4,861	6,175
2010	DIDSON	38,539	119,205	157,744
2011	DIDSON	76,638	197,327	273,965
2012	DIDSON	67,731	130,200	197,931
2013	DIDSON	56,073	171,072	227,145
2014	DIDSON	72,803	141,593	214,396
2015	DIDSON	50,170	112,368	162,538
2016	DIDSON	87,579	166,816	254,395
2017	DIDSON	217,079	266,754	483,833

^a Estimates calculated post season.

^b Incomplete counts, operations stopped before the majority of the run normally passes.

Appendix B. Age, Sex, and Length Data

Table B.1. — Historical age and sex of fall Chum Salmon carcasses sampled on spawning grounds in the Teedriinjik (Chandalar) River, Alaska. Vertebrae were aged by Alaska Department of Fish and Game. Unknown age indicates the number of samples that could not be aged and were not included in age calculations.

Year	Sample size	Unknown age	Age (brood year) n (%)				
2006			<u>0.2 (2003)</u>	<u>0.3 (2002)</u>	<u>0.4 (2001)</u>	<u>0.5 (2000)</u>	<u>0.6 (1999)</u>
Female	72(41.1%)	0 (0.0%)	8 (11.1%)	45 (62.5%)	16 (22.2%)	3 (4.2%)	0 (0.0%)
Male	103(58.9%)	0 (0.0%)	6 (5.8%)	69 (67.0%)	28 (27.2%)	0 (0.0%)	0 (0.0%)
Total	175(100.0%)	0 (0.0%)	14 (8.0%)	114 (65.1%)	44 (25.1%)	3 (1.7%)	0 (0.0%)
2008			<u>0.2 (2005)</u>	<u>0.3 (2004)</u>	<u>0.4 (2003)</u>	<u>0.5 (2002)</u>	<u>0.6 (2001)</u>
Female	102(56.4%)	2 (2.0%)	4 (4.0%)	45 (45.0%)	41 (41.0%)	7 (7.0%)	3 (3.0%)
Male	79(43.6%)	1 (1.3%)	2 (2.6%)	28 (35.9%)	42 (53.8%)	6 (7.7%)	0 (0.0%)
Total	181(100.0%)	3 (1.7%)	6 (3.4%)	73 (41.0%)	83 (46.6%)	13 (7.3%)	3 (1.7%)
2009			<u>0.2 (2006)</u>	<u>0.3 (2005)</u>	<u>0.4 (2004)</u>	<u>0.5 (2003)</u>	<u>0.6 (2002)</u>
Female	104(57.7%)	0 (0.0%)	10 (9.6%)	70 (67.3%)	23 (22.1%)	1 (0.9%)	0 (0.0%)
Male	76(42.2%)	0 (0.0%)	6 (7.9%)	43 (56.6%)	23 (30.3%)	3 (3.9%)	1 (1.3%)
Total	180(100.0%)	0 (0.0%)	16 (8.8%)	113 (62.8%)	46 (25.6%)	4 (2.2%)	1 (0.6%)
2010			<u>0.2 (2007)</u>	<u>0.3 (2006)</u>	<u>0.4 (2005)</u>	<u>0.5 (2004)</u>	<u>0.6 (2003)</u>
Female	124(70.1%)	0 (0.0%)	30 (24.2%)	70 (56.5%)	19 (15.3%)	4 (3.2%)	1 (0.8%)
Male	53(29.9%)	0 (0.0%)	7 (13.2%)	33 (62.3%)	11 (20.8%)	2 (3.8%)	0 (0.0%)
Total	177(100.0%)	0 (0.0%)	37 (20.9%)	103 (58.2%)	30 (16.9%)	6 (3.4%)	1 (0.6%)
2011			<u>0.2 (2008)</u>	<u>0.3 (2007)</u>	<u>0.4 (2006)</u>	<u>0.5 (2005)</u>	<u>0.6 (2004)</u>
Female	277 (51.3%)	6 (2.1%)	4 (1.5%)	161 (61.7%)	92 (35.2%)	14 (5.4%)	0 (0.0%)
Male	263 (48.7%)	3 (1.1%)	3 (1.2%)	116 (44.6%)	126 (48.5%)	15 (5.8%)	0 (0.0%)
Total	540 (100.0%)	9 (1.7%)	7 (1.3%)	277 (52.2%)	218 (41.1%)	29 (5.5%)	0 (0.0%)

Table B.2. — Historical length-at-age of female and male fall Chum Salmon carcasses sampled on Teedriinjik (Chandalar) River spawning grounds, Alaska.

Year	Age	Female					Male				
		N	Mid-eye to fork length (mm)				N	Mid-eye to fork length (mm)			
			Mean	SE	Median	Range		Mean	SE	Median	Range
2006											
	0.2	8	542	13.2	540	480-590	6	573	15.6	585	510-620
	0.3	45	551	3.5	550	500-600	69	583	3.8	580	500-655
	0.4	16	564	5.6	560	530-600	28	604	6.1	600	550-660
	0.5	3	607	18.6	—	570-630	—	—	—	—	—
	0.6	0	—	—	—	—	—	—	—	—	—
	Total	72					103				
2008											
	0.2	4	543	19.3	545	500-580	2	540	10	540	530-550
	0.3	45	552	3.3	550	510-610	28	575	5.9	570	520-640
	0.4	41	578	4.0	580	530-630	42	608	4.3	605	560-700
	0.5	7	560	11.1	560	520-610	6	595	4.3	595	580-610
	0.6	3	593	8.8	590	580-610	0	—	—	—	—
	Total	100					78				
2009											
	0.2	10	553	8.8	555	505-590	6	575	14.1	585	510-610
	0.3	70	557	2.9	558	500-600	43	584	4.3	580	540-650
	0.4	23	565	6.6	570	470-620	23	615	4.8	620	560-660
	0.5	1	590	—	590	—	3	607	16.7	590	590-640
	0.6	0	—	—	—	—	1	660	—	660	—
	Total	104					76				
2010											
	0.2	30	545	4.6	543	490-610	7	599	6.6	600	575-630
	0.3	70	558	3.2	560	500-650	33	605	7.7	610	530-720
	0.4	19	568	8.2	570	500-630	11	586	12.1	580	540-670
	0.5	4	585	11.9	585	560-610	2	595	15.0	595	580-610
	0.6	1	630	—	630	—	0	—	—	—	—
	Total	124					53				
2011											
	0.2	4	531	11.4	530	505-560	3	558	25.9	535	530-610
	0.3	161	562	2.2	560	500-670	116	600	2.9	605	510-660
	0.4	92	582	2.7	580	505-645	126	614	2.3	615	555-695
	0.5	14	594	8.7	588	550-650	15	612	10.3	620	530-695
	0.6	0	—	—	—	—	0	—	—	—	—
	Total	271					260				

Appendix C. Staff Gauge, Water Temperature, and Water Quality Data

Table C.1. — Daily water level (via staff gauge data) and water quality data collected with a YSI sonde at the Teedriinjik (Chandalar) River sonar project, 2017. Daily water quality values are averages of readings taken at 15-minute intervals. Values for pH are not reported due to failure of the pH probe during the season.

Date	Level ^a (m)	Temperature (°C)	Specific conductance (µS/cm)	pH	Turbidity (NTU)	DO	
						mg/L	(% saturation)
5-Aug	2.17						
6-Aug	2.37						
7-Aug	2.46						
8-Aug	2.41						
9-Aug	2.22						
10-Aug	2.09	16.62	407.2		7.3	9.72	99.9
11-Aug	2.00	15.65	400.5		5.2	9.81	98.8
12-Aug	1.98	14.60	391.7		4.0	9.93	97.7
13-Aug	1.95	13.39	383.1		3.7	10.19	97.7
14-Aug	1.93	12.62	376.8		3.0	10.36	97.6
15-Aug	2.28	12.18	354.8		11.7	10.57	98.6
16-Aug	2.69	11.56	285.0		56.8	10.75	98.9
17-Aug	2.82	11.33	275.0		45.0	10.74	98.2
18-Aug	2.82	11.26	285.5		33.9	10.80	98.6
19-Aug	2.86	11.08	304.7		39.5	10.88	99.0
20-Aug	2.87	10.75	330.3		47.0	11.00	99.3
21-Aug	2.86	10.87	340.2		51.2	10.99	99.4
22-Aug	2.82	11.00	346.4		42.4	10.95	99.4
23-Aug	2.76	11.04	345.3		37.8	10.93	99.3
24-Aug	2.78	10.67	340.4		42.8	10.97	98.8
25-Aug	2.74	10.27	336.3		37.1	11.07	98.7
26-Aug	2.65	9.90	337.8		26.1	11.14	98.6
27-Aug	2.61	9.81	338.7		18.5	11.23	99.2
28-Aug	2.63	9.93	331.5		16.1	11.25	99.6
29-Aug	2.66	9.01	324.7		15.8	11.40	98.7
30-Aug	3.06	7.55	308.1		70.6	11.72	98.0
31-Aug	3.36	6.53	343.3		128.5	12.10	98.6
1-Sep	3.49	6.77	343.2		103.4	12.09	99.1
2-Sep	3.34	7.30	344.5		54.9	11.98	99.6
3-Sep	3.17	7.39	345.6		32.7	12.01	100.0
4-Sep	3.16	7.38	334.9		27.7	11.98	99.7
5-Sep	3.08	7.07	317.9		21.9	12.01	99.2
6-Sep	2.97	7.46	318.0		18.1	11.89	99.2
7-Sep	2.87	8.13	318.6		14.7	11.61	98.4
8-Sep	3.00	7.98	316.0		21.6	11.70	98.8
9-Sep	3.18	7.60	322.3		27.5	11.81	98.8
10-Sep	3.18	7.37	340.8		29.9	11.88	98.8
11-Sep	3.11	7.15	367.8		21.8	11.93	98.8
12-Sep	3.00	6.55	372.0		15.6	12.12	98.9
13-Sep	2.91	6.56	369.8		12.2	12.19	99.4
14-Sep	2.87	7.66	361.8		13.3	11.88	99.6
15-Sep	2.92	8.01	345.8		18.2	11.76	99.4
16-Sep	2.93	7.50	347.1		15.4	11.82	98.7
17-Sep	3.04	7.22	375.0		26.6	11.90	98.7
18-Sep	3.04	7.13	394.7		25.2	11.93	98.7
19-Sep	2.92	7.33	400.3		18.2	11.93	99.2
20-Sep	2.81	6.62	396.3		13.7	12.14	99.2
21-Sep	2.72	5.70	389.3		14.4	12.43	99.2
22-Sep	2.64	5.00	382.0		13.7	12.57	98.6
23-Sep	2.56	5.02	370.4		12.7	12.57	98.6
24-Sep	2.49	4.96	357.9		11.7	12.54	98.2
25-Sep	2.42	4.61	353.0		9.7	12.66	98.2
26-Sep	2.33	4.13	349.5		8.7	12.82	98.2
Average ^b	2.74	8.76	347.5		28.6	11.52	98.9
Min ^b	1.93	3.88	271.0		2.4	9.59	96.3
Max ^b	3.49	16.78	409.0		208.4	12.91	100.0

^a Water level data was collected with a staff gage, and has been calibrated to 1989 levels using a benchmark on shore.

^b Average, min, and max values for water level are for the daily values; for the other water data they are for all of the 15-minute readings taken throughout the season.

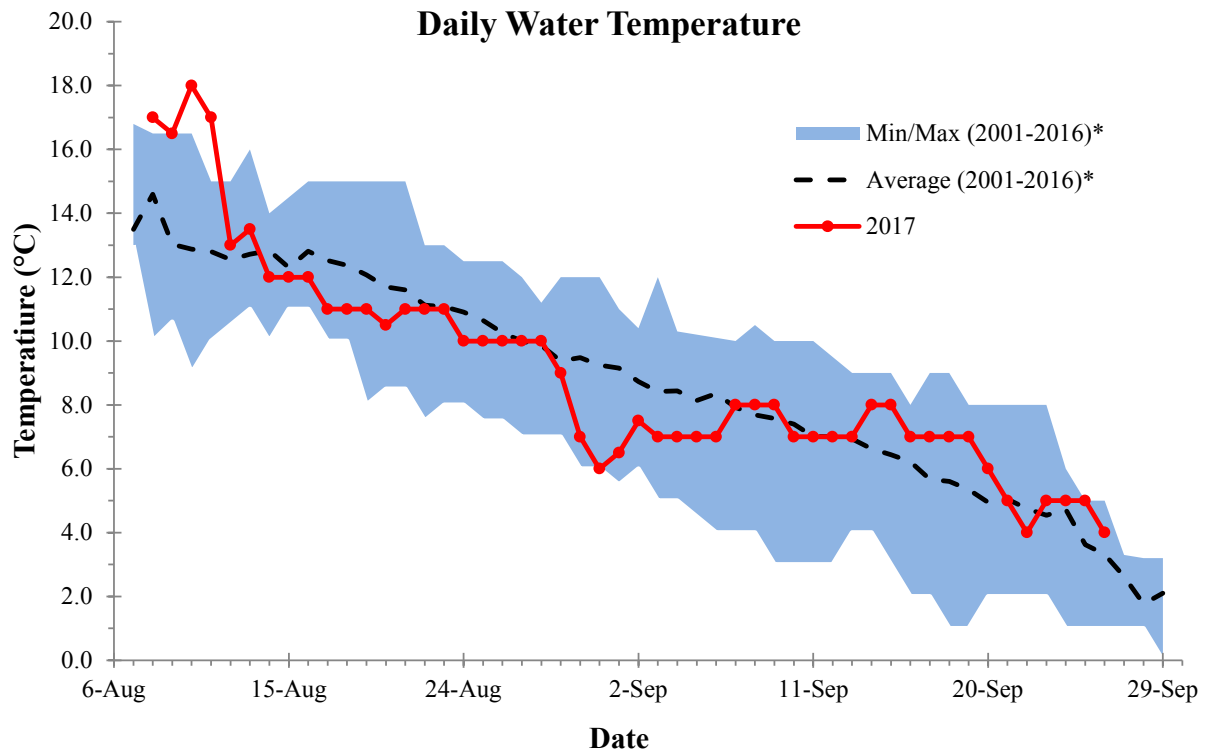


Figure C.1. — Daily water temperatures (collected with glass thermometer) from the Teedriinjik (Chandalar) River sonar site, 2017. Blue shaded area represents historical range of min/max temperatures.

*Historical min/max and averages do not include data from 2003, 2005, 2007, and 2009.

Appendix D. Historical Daily and Cumulative Passage Estimates

Table D.1. — Historical daily and cumulative fall Chum Salmon passage estimates from sonar counts on the Teedriinjik (Chandalar) River. Highlighted cells indicate the mid-point of the run, and boxes indicate the quarter points.

Date	1995		1996		1997		1998		1999	
	Daily	Cum	Daily	Cum	Daily	Cum	Daily	Cum	Daily	Cum
6-Aug										
7-Aug										
8-Aug	1,172	1,172	517	517	619	619	90	90	149	149
9-Aug	928	2,100	341	858	522	1,141	152	242	128	277
10-Aug	861	2,961	323	1,181	682	1,823	215	457	123	400
11-Aug	856	3,817	262	1,443	435	2,258	189	646	119	519
12-Aug	1,269	5,086	356	1,799	752	3,010	162	808	114	633
13-Aug	1,327	6,413	628	2,427	729	3,739	119	927	203	836
14-Aug	1,600	8,013	928	3,355	723	4,462	270	1,197	214	1,050
15-Aug	1,876	9,889	1,209	4,564	838	5,300	395	1,592	368	1,418
16-Aug	1,761	11,650	1,743	6,307	619	5,919	235	1,827	561	1,979
17-Aug	1,672	13,322	2,633	8,940	639	6,558	160	1,987	1,032	3,011
18-Aug	1,741	15,063	3,523	12,463	423	6,981	158	2,145	1,232	4,243
19-Aug	1,851	16,914	4,413	16,876	388	7,369	151	2,296	1,985	6,228
20-Aug	2,297	19,211	5,302	22,178	365	7,734	139	2,435	2,269	8,497
21-Aug	2,729	21,940	6,085	28,263	540	8,274	141	2,576	2,372	10,869
22-Aug	1,988	23,928	6,449	34,712	793	9,067	168	2,744	2,227	13,096
23-Aug	2,596	26,524	7,132	41,844	1,617	10,684	273	3,017	3,266	16,362
24-Aug	6,893	33,417	5,996	47,840	2,263	12,947	318	3,335	3,052	19,414
25-Aug	8,540	41,957	5,165	53,005	3,125	16,072	400	3,735	2,854	22,268
26-Aug	9,666	51,623	6,469	59,474	3,458	19,530	421	4,156	3,679	25,947
27-Aug	6,388	58,011	7,750	67,224	6,103	25,633	486	4,642	3,635	29,582
28-Aug	7,723	65,734	7,572	74,796	5,942	31,575	330	4,972	3,928	33,510
29-Aug	6,842	72,576	6,834	81,630	7,217	38,792	273	5,245	2,961	36,471
30-Aug	8,212	80,788	6,677	88,307	6,661	45,453	651	5,896	2,022	38,493
31-Aug	11,146	91,934	6,737	95,044	6,020	51,473	917	6,813	2,034	40,527
1-Sep	7,229	99,163	7,233	102,277	5,123	56,596	1,230	8,043	1,754	42,281
2-Sep	8,390	107,553	7,982	110,259	4,509	61,105	1,321	9,364	1,974	44,255
3-Sep	8,708	116,261	9,500	119,759	9,720	70,825	1,455	10,819	2,444	46,699
4-Sep	6,136	122,397	7,572	127,331	10,468	81,293	1,379	12,198	2,571	49,270
5-Sep	4,308	126,705	5,837	133,168	13,069	94,362	1,505	13,703	3,716	52,986
6-Sep	3,991	130,696	6,086	139,254	15,951	110,313	1,630	15,333	4,767	57,753
7-Sep	5,354	136,050	6,132	145,386	15,420	125,733	1,675	17,008	3,965	61,718
8-Sep	5,795	141,845	8,090	153,476	12,953	138,686	1,824	18,832	2,775	64,493
9-Sep	3,859	145,704	9,847	163,323	8,872	147,558	2,128	20,960	1,743	66,236
10-Sep	5,087	150,791	9,422	172,745	7,602	155,160	2,429	23,389	1,417	67,653
11-Sep	3,825	154,616	9,870	182,615	5,458	160,618	2,503	25,892	1,227	68,880
12-Sep	3,728	158,344	9,263	191,878	4,660	165,278	2,512	28,404	1,195	70,075
13-Sep	5,764	164,108	10,708	202,586	4,109	169,387	2,723	31,127	1,238	71,313
14-Sep	3,672	167,780	10,095	212,681	3,956	173,343	2,524	33,651	1,363	72,676
15-Sep	3,739	171,519	9,527	222,208	3,900	177,243	2,273	35,924	1,133	73,809
16-Sep	6,104	177,623	8,324	230,532	4,124	181,367	2,747	38,671	1,357	75,166
17-Sep	7,063	184,686	8,439	238,971	4,264	185,631	4,999	43,670	1,340	76,506
18-Sep	5,089	189,775	8,274	247,245	3,656	189,287	5,935	49,605	1,352	77,858
19-Sep	5,819	195,594	8,086	255,331	3,513	192,800	4,731	54,336	1,332	79,190
20-Sep	4,186	199,780	7,836	263,167	2,320	195,120	4,401	58,737	1,510	80,700
21-Sep	4,086	203,866	9,605	272,772	2,428	197,548	4,053	62,790	1,324	82,024
22-Sep	4,304	208,170	8,227	280,999	2,326	199,874	3,329	66,119	1,628	83,652
23-Sep							2,738	68,857	1,490	85,142
24-Sep							2,498	71,355	1,362	86,504
25-Sep							2,336	73,691	1,112	87,616
26-Sep							2,103	75,794	1,046	88,662
27-Sep										
28-Sep										
29-Sep										

Table D.1. — Continued.

Date	2000		2001		2002		2003		2004	
	Daily	Cum	Daily	Cum	Daily	Cum	Daily	Cum	Daily	Cum
6-Aug										
7-Aug										
8-Aug	226	226	454	454	216	216	310	310	880	880
9-Aug	232	458	368	822	665	881	395	705	907	1,787
10-Aug	222	680	355	1,177	774	1,655	449	1,154	995	2,782
11-Aug	260	940	317	1,494	600	2,255	872	2,026	991	3,773
12-Aug	200	1,140	385	1,879	905	3,160	894	2,920	1,077	4,850
13-Aug	238	1,378	322	2,201	569	3,729	792	3,712	1,031	5,881
14-Aug	264	1,642	626	2,827	270	3,999	1,193	4,905	921	6,802
15-Aug	216	1,858	969	3,796	623	4,622	1,598	6,503	888	7,690
16-Aug	240	2,098	1,270	5,066	691	5,313	1,980	8,483	1,016	8,706
17-Aug	500	2,598	1,561	6,627	772	6,085	3,551	12,035	1,193	9,899
18-Aug	451	3,049	7,024	13,651	641	6,726	3,747	15,781	1,350	11,249
19-Aug	460	3,509	5,108	18,759	959	7,685	3,294	19,076	1,374	12,623
20-Aug	665	4,174	3,164	21,923	683	8,368	3,015	22,091	1,610	14,233
21-Aug	621	4,795	2,576	24,499	469	8,837	4,363	26,454	1,488	15,721
22-Aug	706	5,501	2,279	26,778	481	9,318	5,789	32,243	1,230	16,951
23-Aug	591	6,092	2,902	29,680	604	9,922	6,427	38,671	1,555	18,506
24-Aug	2,270	8,362	2,744	32,424	700	10,622	5,237	43,908	981	19,487
25-Aug	1,616	9,978	2,630	35,054	721	11,343	4,537	48,445	787	20,274
26-Aug	1,231	11,209	2,272	37,326	1,074	12,417	3,992	52,436	699	20,973
27-Aug	1,051	12,260	2,282	39,608	1,260	13,677	5,073	57,509	738	21,711
28-Aug	1,742	14,002	1,940	41,548	1,644	15,321	6,170	63,680	1,602	23,313
29-Aug	1,598	15,600	2,728	44,276	2,230	17,551	7,896	71,576	2,485	25,798
30-Aug	1,303	16,903	2,066	46,342	1,722	19,273	7,980	79,556	2,622	28,420
31-Aug	1,943	18,846	2,359	48,701	2,790	22,063	7,828	87,384	3,985	32,405
1-Sep	2,601	21,447	2,307	51,008	2,541	24,604	7,639	95,023	5,247	37,652
2-Sep	1,981	23,428	2,575	53,583	2,281	26,885	6,812	101,834	4,910	42,562
3-Sep	2,021	25,449	2,478	56,061	1,977	28,862	7,357	109,191	5,953	48,515
4-Sep	2,159	27,608	3,421	59,482	2,038	30,900	10,955	120,146	7,167	55,682
5-Sep	2,150	29,758	3,540	63,022	1,389	32,289	8,978	129,124	4,438	60,120
6-Sep	2,262	32,020	3,086	66,108	1,458	33,747	7,050	136,174	5,357	65,477
7-Sep	1,902	33,922	4,437	70,545	1,530	35,277	4,667	140,842	6,344	71,821
8-Sep	1,983	35,905	3,860	74,405	1,780	37,057	3,387	144,229	6,053	77,874
9-Sep	1,650	37,555	3,746	78,151	1,857	38,914	3,899	148,127	5,308	83,182
10-Sep	1,791	39,346	4,176	82,327	1,981	40,895	5,659	153,786	4,473	87,655
11-Sep	1,921	41,267	3,108	85,435	2,922	43,817	4,856	158,642	5,415	93,070
12-Sep	1,484	42,751	3,311	88,746	2,830	46,647	4,329	162,972	5,491	98,561
13-Sep	1,496	44,247	3,107	91,853	3,410	50,057	3,954	166,926	6,525	105,086
14-Sep	1,517	45,764	2,320	94,173	4,112	54,169	3,795	170,721	5,741	110,827
15-Sep	1,160	46,924	2,208	96,381	4,145	58,314	4,520	175,241	4,055	114,882
16-Sep	1,292	48,216	2,165	98,546	4,152	62,466	4,789	180,030	2,515	117,397
17-Sep	1,225	49,441	2,173	100,719	3,671	66,137	6,049	186,079	1,669	119,066
18-Sep	1,409	50,850	1,696	102,415	4,033	70,170	3,565	189,644	2,280	121,346
19-Sep	1,289	52,139	1,525	103,940	3,490	73,660	2,307	191,951	2,731	124,077
20-Sep	1,690	53,829	1,530	105,470	3,356	77,016	3,592	195,543	2,765	126,842
21-Sep	1,765	55,594	1,293	106,763	2,846	79,862	5,551	201,094	3,401	130,243
22-Sep	1,607	57,201	1,203	107,966	2,174	82,036	3,430	204,524	6,845	137,088
23-Sep	1,113	58,314	1,201	109,167	2,077	84,113	3,047	207,571		
24-Sep	1,280	59,594	786	109,953	2,095	86,208	2,466	210,037		
25-Sep	1,665	61,259	578	110,531	1,904	88,112	2,590	212,627		
26-Sep	1,340	62,599	440	110,971	1,735	89,847	1,801	214,428		
27-Sep										
28-Sep										
29-Sep										

Table D.1. — Continued, (no data for 2009, when the project was terminated early, before the most of the run occurred).

Date	2005		2006		2007		2008		2010	
	Daily	Cum	Daily	Cum	Daily	Cum	Daily	Cum	Daily	Cum
6-Aug										
7-Aug										
8-Aug	2,819	2,819	570	570	269	269	521	521	173	173
9-Aug	4,117	6,936	526	1,096	375	644	673	1,194	130	303
10-Aug	5,235	12,171	625	1,721	551	1,195	717	1,911	258	561
11-Aug	5,899	18,070	589	2,310	553	1,748	662	2,573	386	947
12-Aug	5,214	23,284	751	3,061	628	2,376	877	3,450	514	1,461
13-Aug	5,972	29,256	871	3,932	504	2,880	1,332	4,782	641	2,102
14-Aug	6,252	35,508	1,074	5,006	522	3,402	1,008	5,790	769	2,871
15-Aug	5,923	41,431	2,960	7,966	553	3,955	1,136	6,926	897	3,768
16-Aug	6,893	48,324	1,785	9,751	572	4,527	1,054	7,980	1,025	4,793
17-Aug	7,154	55,478	1,082	10,833	674	5,201	1,321	9,301	1,152	5,945
18-Aug	5,245	60,723	1,276	12,109	786	5,987	1,099	10,400	1,381	7,326
19-Aug	6,233	66,956	1,646	13,755	591	6,578	862	11,262	1,562	8,888
20-Aug	5,820	72,776	1,931	15,686	496	7,074	755	12,017	1,865	10,753
21-Aug	6,479	79,255	2,216	17,902	454	7,528	967	12,984	1,468	12,221
22-Aug	5,303	84,558	2,501	20,403	437	7,965	819	13,803	1,596	13,817
23-Aug	5,217	89,775	2,786	23,189	419	8,384	939	14,742	1,509	15,326
24-Aug	4,495	94,270	3,071	26,260	427	8,811	1,006	15,748	1,893	17,219
25-Aug	4,707	98,977	3,356	29,616	408	9,219	1,158	16,906	2,096	19,315
26-Aug	3,572	102,549	3,641	33,257	336	9,555	1,799	18,705	2,179	21,494
27-Aug	4,798	107,347	3,926	37,183	381	9,936	2,318	21,023	2,055	23,549
28-Aug	5,510	112,857	4,501	41,684	417	10,353	2,424	23,447	2,310	25,859
29-Aug	6,186	119,043	6,160	47,844	458	10,811	4,259	27,706	2,392	28,251
30-Aug	8,162	127,205	8,420	56,264	476	11,287	4,596	32,302	1,926	30,177
31-Aug	7,608	134,813	11,266	67,530	556	11,843	5,376	37,678	2,046	32,223
1-Sep	18,372	153,185	11,041	78,571	897	12,740	6,184	43,862	1,937	34,160
2-Sep	12,774	165,959	11,815	90,386	994	13,734	6,440	50,302	1,883	36,043
3-Sep	17,290	183,249	10,819	101,205	1,658	15,392	7,210	57,512	1,847	37,890
4-Sep	23,630	206,879	9,762	110,967	2,965	18,357	8,411	65,923	1,816	39,706
5-Sep	25,251	232,130	7,091	118,058	5,086	23,443	7,530	73,453	1,914	41,620
6-Sep	24,374	256,504	6,522	124,580	6,739	30,182	6,979	80,432	2,330	43,950
7-Sep	22,788	279,292	5,744	130,324	9,676	39,858	6,814	87,246	3,224	47,174
8-Sep	22,831	302,123	5,675	135,999	13,137	52,995	5,439	92,685	4,058	51,232
9-Sep	18,256	320,379	6,336	142,335	14,952	67,947	4,535	97,220	4,501	55,733
10-Sep	12,488	332,867	5,886	148,221	14,571	82,518	3,982	101,202	5,183	60,916
11-Sep	16,035	348,902	6,569	154,790	17,754	100,272	3,624	104,826	6,330	67,246
12-Sep	17,056	365,958	6,412	161,202	17,067	117,339	3,765	108,591	7,344	74,590
13-Sep	12,242	378,200	7,176	168,378	15,931	133,270	3,501	112,092	8,106	82,696
14-Sep	12,973	391,173	8,324	176,702	16,398	149,668	3,189	115,281	8,103	90,799
15-Sep	11,966	403,139	8,440	185,142	13,399	163,067	2,851	118,132	8,255	99,054
16-Sep	8,848	411,987	8,721	193,863	12,772	175,839	3,215	121,347	7,820	106,874
17-Sep	8,511	420,498	8,082	201,945	11,374	187,213	3,626	124,973	8,160	115,034
18-Sep	9,271	429,769	8,499	210,444	6,934	194,147	4,107	129,080	7,028	122,062
19-Sep	9,435	439,204	6,805	217,249	5,690	199,837	4,085	133,165	6,991	129,053
20-Sep	8,485	447,689	6,362	223,611	4,644	204,481	5,082	138,247	6,538	135,591
21-Sep	6,875	454,564	4,977	228,588	3,598	208,079	4,008	142,255	6,154	141,745
22-Sep	9,396	463,960	3,931	232,519	3,364	211,443	4,108	146,363	4,459	146,204
23-Sep	8,033	471,993	3,997	236,516	4,102	215,545	3,660	150,023	3,337	149,541
24-Sep	9,513	481,506	3,315	239,831	4,099	219,644	4,145	154,168	2,804	152,345
25-Sep	7,086	488,592	2,740	242,571	4,316	223,960	3,630	157,798	2,854	155,199
26-Sep	7,892	496,484	2,519	245,090	4,095	228,055	4,226	162,024	2,545	157,744
27-Sep										
28-Sep										
29-Sep										

Table D.1. — Continued.

Date	2011		2012		2013		2014		2015	
	Daily	Cum	Daily	Cum	Daily	Cum	Daily	Cum	Daily	Cum
6-Aug							262	262		
7-Aug							222	484		
8-Aug			670	670	1,134	1,134	226	710	272	272
9-Aug	481	481	669	1,339	1,053	2,187	302	1,012	282	554
10-Aug	760	1,241	673	2,012	1,028	3,215	402	1,414	328	882
11-Aug	835	2,076	659	2,671	1,030	4,245	484	1,898	190	1,072
12-Aug	840	2,916	642	3,313	954	5,199	878	2,776	280	1,352
13-Aug	813	3,729	640	3,953	1,072	6,271	1,156	3,932	444	1,796
14-Aug	729	4,458	788	4,741	1,093	7,364	1,472	5,404	600	2,396
15-Aug	693	5,151	888	5,629	991	8,355	2,470	7,874	898	3,294
16-Aug	786	5,937	988	6,617	1,012	9,367	2,486	10,360	1,014	4,308
17-Aug	1,153	7,090	1,215	7,832	872	10,239	3,884	14,244	1,110	5,418
18-Aug	1,349	8,439	1,095	8,927	1,210	11,449	4,706	18,950	1,368	6,786
19-Aug	1,819	10,258	1,392	10,319	1,343	12,792	4,054	23,004	1,752	8,538
20-Aug	1,638	11,896	1,731	12,050	1,341	14,133	5,030	28,034	2,307	10,845
21-Aug	1,081	12,977	1,998	14,048	1,025	15,158	5,490	33,524	2,356	13,201
22-Aug	1,237	14,214	2,114	16,162	2,000	17,158	5,150	38,674	2,405	15,606
23-Aug	1,384	15,598	3,000	19,162	2,463	19,621	4,914	43,588	2,454	18,060
24-Aug	1,331	16,929	4,049	23,211	2,351	21,972	3,996	47,584	2,502	20,562
25-Aug	1,360	18,289	3,859	27,070	1,920	23,892	3,680	51,264	2,551	23,113
26-Aug	1,840	20,129	3,489	30,559	1,642	25,534	3,562	54,826	2,600	25,713
27-Aug	2,971	23,100	4,104	34,663	1,740	27,274	3,304	58,130	2,649	28,362
28-Aug	4,527	27,627	4,410	39,073	2,332	29,606	2,662	60,792	2,697	31,059
29-Aug	5,985	33,612	4,353	43,426	2,957	32,563	3,006	63,798	2,747	33,806
30-Aug	7,672	41,284	5,924	49,350	3,917	36,480	2,443	66,241	2,795	36,601
31-Aug	9,218	50,502	7,410	56,760	2,573	39,053	2,152	68,393	2,844	39,445
1-Sep	9,918	60,420	5,734	62,494	2,749	41,802	2,620	71,013	2,892	42,337
2-Sep	10,228	70,648	5,221	67,715	2,918	44,720	3,610	74,623	2,942	45,279
3-Sep	11,965	82,613	5,040	72,755	3,303	48,023	2,986	77,609	2,990	48,269
4-Sep	11,836	94,449	5,379	78,134	3,209	51,232	2,368	79,977	2,909	51,178
5-Sep	11,185	105,634	4,316	82,450	3,881	55,113	1,750	81,727	2,802	53,980
6-Sep	10,787	116,421	4,012	86,462	3,395	58,508	2,232	83,959	3,122	57,102
7-Sep	7,711	124,132	3,123	89,585	3,783	62,291	1,850	85,809	3,216	60,318
8-Sep	9,406	133,538	3,043	92,628	4,333	66,624	1,894	87,703	3,661	63,979
9-Sep	10,524	144,062	2,963	95,591	5,612	72,236	2,752	90,455	4,368	68,347
10-Sep	8,010	152,072	2,882	98,473	7,506	79,742	3,544	93,999	3,760	72,107
11-Sep	6,554	158,626	2,802	101,275	9,069	88,811	3,728	97,727	4,162	76,269
12-Sep	6,809	165,435	2,984	104,259	11,157	99,968	4,253	101,980	4,490	80,759
13-Sep	7,486	172,921	3,297	107,556	12,687	112,655	5,276	107,256	5,656	86,415
14-Sep	7,132	180,053	4,463	112,019	11,042	123,697	6,438	113,694	5,082	91,497
15-Sep	7,458	187,511	4,843	116,862	10,104	133,801	7,722	121,416	5,958	97,455
16-Sep	7,256	194,767	6,006	122,868	10,600	144,401	10,222	131,638	7,787	105,242
17-Sep	8,123	202,890	9,631	132,499	8,588	152,989	9,100	140,738	6,615	111,857
18-Sep	7,914	210,804	8,659	141,158	7,530	160,519	9,692	150,430	6,670	118,527
19-Sep	8,773	219,577	7,093	148,251	10,424	170,943	9,564	159,994	7,024	125,551
20-Sep	8,789	228,366	8,000	156,251	10,184	181,127	9,640	169,634	6,754	132,305
21-Sep	7,772	236,138	8,643	164,894	6,280	187,407	9,038	178,672	5,630	137,935
22-Sep	8,487	244,625	6,220	171,114	7,598	195,005	7,974	186,646	3,764	141,699
23-Sep	8,395	253,020	4,418	175,532	8,735	203,740	7,516	194,162	4,778	146,477
24-Sep	7,369	260,389	5,642	181,174	8,405	212,145	5,682	199,844	5,576	152,053
25-Sep	7,269	267,658	4,037	185,211	8,353	220,498	4,174	204,018	4,304	156,357
26-Sep	6,307	273,965	3,411	188,622	6,647	227,145	4,094	208,112	3,562	159,919
27-Sep			3,273	191,895			4,076	212,188	2,619	162,538
28-Sep			3,278	195,173			2,208	214,396		
29-Sep			2,758	197,931						

Table D.1. — Continued.

Date	2016		2017	
	Daily	Cum	Daily	Cum
6-Aug				
7-Aug				
8-Aug	443	443	1,706	1,706
9-Aug	471	914	1,356	3,062
10-Aug	538	1,452	987	4,049
11-Aug	602	2,054	692	4,741
12-Aug	481	2,535	594	5,335
13-Aug	427	2,962	520	5,855
14-Aug	649	3,611	642	6,497
15-Aug	609	4,220	956	7,453
16-Aug	832	5,052	1,485	8,938
17-Aug	1,443	6,495	2,339	11,277
18-Aug	1,776	8,271	2,462	13,739
19-Aug	2,282	10,553	2,874	16,613
20-Aug	3,147	13,700	2,942	19,555
21-Aug	4,033	17,733	2,744	22,299
22-Aug	4,608	22,341	2,920	25,219
23-Aug	4,856	27,197	3,735	28,954
24-Aug	5,839	33,036	3,866	32,820
25-Aug	5,544	38,580	4,000	36,820
26-Aug	4,752	43,332	4,821	41,641
27-Aug	5,424	48,756	5,650	47,291
28-Aug	5,293	54,049	6,738	54,029
29-Aug	5,066	59,115	6,230	60,259
30-Aug	4,824	63,939	5,998	66,257
31-Aug	4,906	68,845	6,191	72,448
1-Sep	4,973	73,818	6,384	78,831
2-Sep	5,063	78,881	6,576	85,408
3-Sep	4,526	83,407	6,769	92,177
4-Sep	4,545	87,952	6,268	98,445
5-Sep	4,740	92,692	5,764	104,209
6-Sep	4,403	97,095	5,328	109,537
7-Sep	3,258	100,353	5,592	115,129
8-Sep	2,929	103,282	7,216	122,345
9-Sep	3,045	106,327	10,330	132,675
10-Sep	3,082	109,409	12,012	144,687
11-Sep	3,167	112,576	16,642	161,329
12-Sep	3,622	116,198	20,334	181,663
13-Sep	3,247	119,445	23,388	205,051
14-Sep	3,387	122,832	31,542	236,593
15-Sep	4,198	127,030	34,214	270,807
16-Sep	5,750	132,780	33,240	304,047
17-Sep	8,299	141,079	27,584	331,631
18-Sep	11,136	152,215	25,838	357,469
19-Sep	11,528	163,743	24,064	381,533
20-Sep	12,737	176,480	20,886	402,419
21-Sep	13,311	189,791	21,042	423,461
22-Sep	14,449	204,240	18,942	442,403
23-Sep	14,106	218,346	13,469	455,872
24-Sep	12,033	230,379	11,380	467,252
25-Sep	11,413	241,792	8,421	475,673
26-Sep	12,603	254,395	8,160	483,833
27-Sep				
28-Sep				
29-Sep				